

RicoSLAM: Long-Term Localization and Mapping in Dynamic Environments

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THESIS RESEARCH PLAN



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Resumo

PUT HERE ABSTRACT IN PORTUGUESE.

Palavras-chave: localização e mapeamento simultâneo, robô móvel autônomo, autonomia a longo prazo, ambientes dinâmicos.

Abstract

PUT HERE ABSTRACT IN PORTUGUESE.

Keywords: Simultaneous Localization and Mapping (SLAM), Autonomous Mobile Robot (AMR), long-term autonomy, lifelong SLAM, dynamic environments.

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Abbreviations

AI Artificial Intelligence. 3

AMR Autonomous Mobile Robot. 1–4, 10, 22, 23

ARMA Auto-regressive Moving Average Model. 3, 21, 23

ATE Absolute Trajectory Error. 17, 18, 20, 21

BDVA Big Data Value Association. 3

BoW Bag-of-Words. 20

CAD Computer-Aided Design. 18

CAGR Compound Annual Growth Rate. 1

CLAIRE Confederation of Laboratories for Artificial Intelligence Research in Europe. 3

CML Concurrent Mapping and Localization. 10

CNN Convolutional Neural Network. 2, 20, 23

DBN Dynamic Bayesian Network. 13

DEL Dynamic Edge Link. 19

DFR-SLAM Dynamic visual Feature Removal Simultaneous Localization and Mapping. 18

DSRM Design Science Research Methodology. 4, 5

EKF Extended Kalman Filter. 13, 14

ELLIS European Lab for Learning & Intelligent Systems. 3

EurAI European Association for Artificial Intelligence. 3

euRobotics European Robotics. 3

FreMEn Frequency Map Enhancement. 3, 21, 23

FSH Feature Stability Histogram. 17

HMM Hidden Markov Model. 21

HOG Histogram of Oriented Gradients. 21

ICP Iterative Closest Point. 12, 18, 19, 21

iSAM incremental Smoothing And Mapping). 15

LC-CRF Long-term Consistent Conditional Random Field. 18

LiDAR Light Detection And Ranging. 2, 12, 18–20, 23

LTD Long-Term Dynamics. 21

LTM Long-Term Memory. 3, 17, 20, 22

MCL Monte Carlo Localization. 17, 18, 21, 22

NDT Normal Distribution Transform. 12, 19, 22

NICP Normal Iterative Closest Point. 12

POV Point-Of-View. 4

- PRISMA** Preferred Reporting Items for Systematic reviews and Meta-Analysis. 7–9, 16
- RBPF** Rao-Blackwellized Particle Filter. 13
- RMSE** Root Mean Square Error. 21
- RNN** Recurrent Neural Network. 20
- ROS** Robot Operating System. 15
- SDF** Signed Distance Field. 18
- SDG** Sustainable Development Goal. 4
- SGC-VSLAM** Semantic and Geometric Constraints Visual Simultaneous Localization and Mapping. 20, 21
- SIVO** Semantically Informed Visual Odometry. 20
- SLAM** Simultaneous Localization and Mapping. 1–5, 7–16, 21–23
- SM** Sensory Memory. 16, 17
- SRIDA** Strategic Research, Innovation and Deployment Agenda. 3, 4
- STD** Short-Term Dynamics. 21
- STM** Short-Term Memory. 3, 17, 22
- TORO** Tree-based netwORk Optimizer. 15

Chapter 1

Introduction

This research aims to contribute to the progress of the current state of the art on **Simultaneous Localization and Mapping (SLAM)**. This work focuses on challenges inherent to the long-term operation of **Autonomous Mobile Robots (AMRs)** while performing SLAM in dynamic indoor environments. The main goals are to maintain a stable and up-to-date representation of the environment and, consequently, improve the localization performance of **AMRs**. In this context, the proposed approach should develop new algorithms and take advantage of emerging technologies to overcome the challenges found through the research work.

This chapter is divided into five sections. Section 1.1 contextualizes this research work and describes the primary challenges related to long-term **Simultaneous Localization and Mapping (SLAM)**. Section 1.2 presents the motivations that led to the formulation of this research plan. Then, Section 1.3 establishes the goals for the PhD thesis. Section 1.4 details the research methodology that will support the research work. Finally, Section 1.5 defines the structure of this document.

1.1 Context

Nowadays, logistic and retail companies, warehouses, healthcare institutions, automotive and aerospace industries, agriculture businesses, and healthcare institutions are using **AMRs** to automate tasks such as transportation and security monitoring. The tasks automation improves operational efficiency, flexibility, and productivity achieving higher efficiencies and lower production costs [1]. Consequently, these improvements contribute to the growth of the global economy and increase the global interest in the **AMR** market. Indeed, a study elaborated by AgileIntel Research [2] estimates the value of the global **AMR** market at US\$2,525.66 million in 2021 and is estimated to grow at a **Compound Annual Growth Rate (CAGR)** of 15.4% between 2022 and 2029, as illustrated in Figure 1.1. The conclusions of the study highlight the importance of **AMRs** for global economics while also forecast an increasing demand for **AMR** systems in the following years from 2022 [2].

An **AMR** requires a representation of its surroundings by extracting meaningful data from its sensors to localize itself relative to the environment [3]. **SLAM** addresses this problem by incorporating the robot state estimation concurrently with the mapping process to build a representation of the environment perceived by the robot. The state is described by the robot's pose (position and orientation), even though other variables may be included in the state, such as velocity and calibration parameters. As for the environment representation, the mapping process originates a map incrementally built when exploring unknown areas or refined on passages through known locations in order to describe the environment in which the robot operates [4].

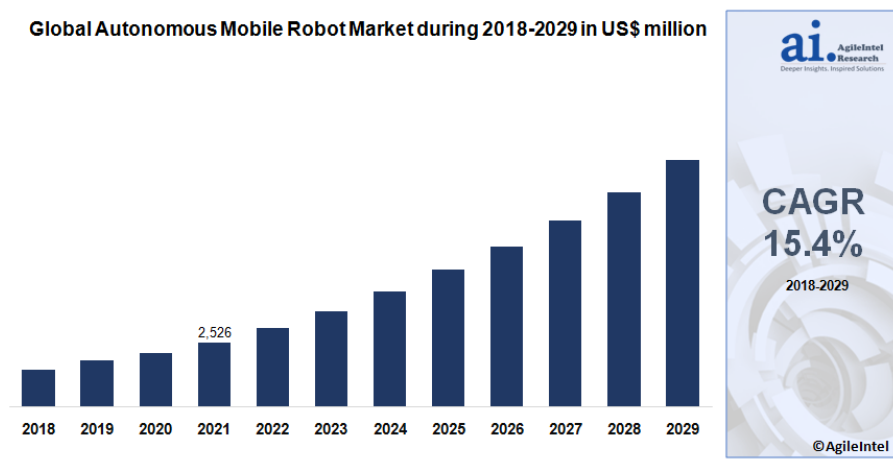


Figure 1.1: Global **Autonomous Mobile Robot (AMR)** market estimated growth by AgileIntel Research during 2018–2029 in US\$ million [2].

In a static scene, the robot would only need to map once because the map would remain consistent with the environment throughout time. However, industrial locations, agricultural sites, warehouses, and service-oriented applications such as shopping centers and healthcare institutions have inherently moving elements in the scene (humans, objects), reconfiguration of the environment (reorganization of machines, patient beds), and appearance variations (lighting, weather, seasonal, or daytime changes). The deployment of **AMRs** in these varying conditions are a challenge for the **SLAM** system, where the system should decide how (consider the most current state, the evolution of the environment dynamics, only the most static changes?) and when to update the map (when the variations occur, after a certain time?). This challenge is also known as the stability-plasticity dilemma, where a lifelong **SLAM** system should both adapt to new environment changes and preserve old states over time [5]. Furthermore, the map would grow indefinitely when gathering new information from the environment without any map management system. This ever-growing problem poses another challenge for the **SLAM** system in the long-term due to the limited computational resources of the mapping vehicle. Indeed, the map size should be dependent on the environment area and not on the trajectory length of the mapping process nor operation time, only growing when the robot would explore unknown locations [6].

Currently, the literature is predominantly focused on the appearance invariance challenge of long-term localization and mapping, especially in outdoor environments, given the recent growing interest in autonomous vehicles. Although vision sensors are greatly affected by appearance variations, **Convolutional Neural Network (CNN)**-based features extracted from the visual images are being used for localization in challenging outdoor scenes, given their more discriminative properties compared to handcrafted features [7–12]. The current research also uses other sensor modalities, such as **Light Detection And Ranging (LiDAR)** [13–16] (also known as 3D laser scanners) and Radar [16, 17], to overcome some limitations of vision sensors. While visual data is affected by illumination changes of day-night situations or changing weather and seasonal conditions, laser-based localization does not suffer from illumination variance. Even so, fog or direct light can still affect the laser readings. As for radar, the sensor is invariant to both lighting and weather changes. Still, noisy measurements affect the performance of radar-based **SLAM** in long-term scenarios [16]. Moreover, the existent literature also evaluates the long-term stability of features to only using stable ones for localization and mapping, using as scoring functions predictors such as number of observations and concentration ratio [18, 19].

Several works already focus on the ever-growing problem of continuously performing **SLAM**. The concept of mutual information from information theory is the most common methodology to determine which nodes to remove from the map for the graph formulation of **SLAM** [6, 20–23]. However, this methodology is mostly intended for offline execution (*e.g.*, between running sessions) due to its additional computational cost when operating online. Other methodologies to manage the size of the graph evaluate outdated information between current view and the map [24, 25], the spatial density of nodes [26, 27], and the node time recency [28, 29]. In feature-based **SLAM**, management policies such as counting matching observations of features [30, 31] and using multiple predictor functions [18, 19, 32, 33], similarly to appearance invariance methods, also try to constraint the number of mapped features to only be dependent on the explored environment area.

As for dealing with dynamic environments, an intuitive approach is to distinguish between static and dynamic observations, favoring the stable ones for localization, and having two maps representing the most stable and the more dynamic representations of the environment [24, 34, 35]. Inspired by the human memory multi-store model of Atkinson and Shiffrin [36], **Short-Term Memory (STM)** and **Long-Term Memory (LTM)** maps try to capture at least two different levels of dynamics [5, 30, 31]. Furthermore, the rejection of dynamic objects or features for localization and mapping to only consider stable observations is performed in the literature by methods such as the YOLO network [35, 37] and pixel-wise semantic segmentation [38–40]. The methodologies mentioned previously for dealing with dynamic environments model the environment dynamics as static (permanent change), semi-static (short-term static object or static at the current observation instant), or dynamic (moving object in the scene). In contrast, assuming the independence of observations, Markov-based processes can be used to model and predict a value for the environment dynamics [41, 42]. However, these models are usually intended for offline usage between sessions and the assumption of independent observations may not be valid because static and dynamic objects may influence the dynamics of their surroundings. Although the **Frequency Map Enhancement (FreME)** [43] algorithm estimates online the dynamics through spectral analysis, the method only assumes periodic changes in the environment. The **Auto-regressive Moving Average Model (ARMA)** [44] models both aperiodic and periodic changes but, still, its offline operation does not allow an online estimation of the environment dynamics.

1.2 Motivation

A joint partnership between major European activities for **Artificial Intelligence (AI)** – **Confederation of Laboratories for Artificial Intelligence Research in Europe (CLAIRE)**, **European Lab for Learning & Intelligent Systems (ELLIS)**, and **European Association for Artificial Intelligence (EurAI)** –, **Data – Big Data Value Association (BDVA)** –, and **Robotics – European Robotics (euRobotics)** – released a **Strategic Research, Innovation and Deployment Agenda (SRIDA)** [45] in September 2020. This agenda identifies key and critical technical areas for the future of Robotics, among them are dealing with unpredictable and unknown environments and irreversible actions. Indeed, in many service applications, surrounding environments of autonomous systems usually involve the interaction with humans. Furthermore, bad policies can bring autonomous systems to unrecoverable states from which learning is no longer possible, requiring the environment to be reset to start the process again. For example, the influence of dynamic elements in the process of **SLAM** with **AMRs** can make the environment representation saved in the robot outdated and unusable due to changes in the scene without any module to update the map accordingly to the environment changes. An outdated map representation would

required a specialized operator to remap the environment, possibly redefining the trajectories if needed, and then restart the autonomous system operation.

Subsequently, the interpretation and processing of dynamic changes in the environment by **AMRs** would enable the maintenance of an up-to-date map representation of the robots' surroundings. Robot systems would be able to better understand their operating environment, which may include humans, while also improving the efficiency and safety of these systems. For these outcomes, **SRIDA** [45] also identifies technology and perception enablers such as robot systems being able to interpret sensor data in real-time – particularly in complex environments or fusion of multi-modal sensor data (vision, ranging, acoustic) to increase self-awareness of those systems –, matching local **Point-Of-View (POV)** with world views, and full 3D perception systems and sensors able to decompose and interpret whole scenes in real-time to 4D.

1.3 Goals

The main scope of this research work, *RicoSLAM: Long-Term Localization and Mapping in Dynamic Environments*, is performing lifelong **SLAM** within dynamic environments with an **AMR**. The focus is developing map maintenance algorithms capable of handling dynamic changes and moving elements in the operating area of autonomous robot systems. Thus, the main goals of this research proposal aiming to contribute to the progress of the current state of the art on **SLAM** are the following ones:

- modeling of environment dynamics perceived by multi-modal sensors (2D and 3D ranging data, vision) in the map representation;
- account the estimated values of the environment dynamics incorporated in the map for both localization and mapping processes;
- achieve long-term localization and mapping in dynamic environments.

Within the scope of the 2030 Agenda for sustainable development agreed by 195 nations in September 2015 [46], this research work falls under the following **Sustainable Development Goal (SDG)**:

- SDG8: Decent work and economic growth;
- SDG9: Industry, innovation and infrastructure;
- SDG12: Responsible Consumption and Production.

1.4 Research methodology

The definition of a base research methodology is crucial to formulate a robust research strategy for obtaining successful results. In engineering disciplines, the **Design Science Research Methodology (DSRM)** is considered a valid and valuable method to obtain applicable and effective problem solutions incrementally [47]. Thus, this PhD thesis will follow a **DSRM** strategy as the one presented in Peffers *et. al.* [47] and illustrated in Figure 1.2. **DSRM** incorporates the principles, practices, and procedures to carry out research projects. The methodology consists on the following nominal sequential processes:

1. Problem identification and motivation;
2. Definition of the objectives for a solution;
3. Design and development;
4. Demonstration;
5. Evaluation;

6. Communication.

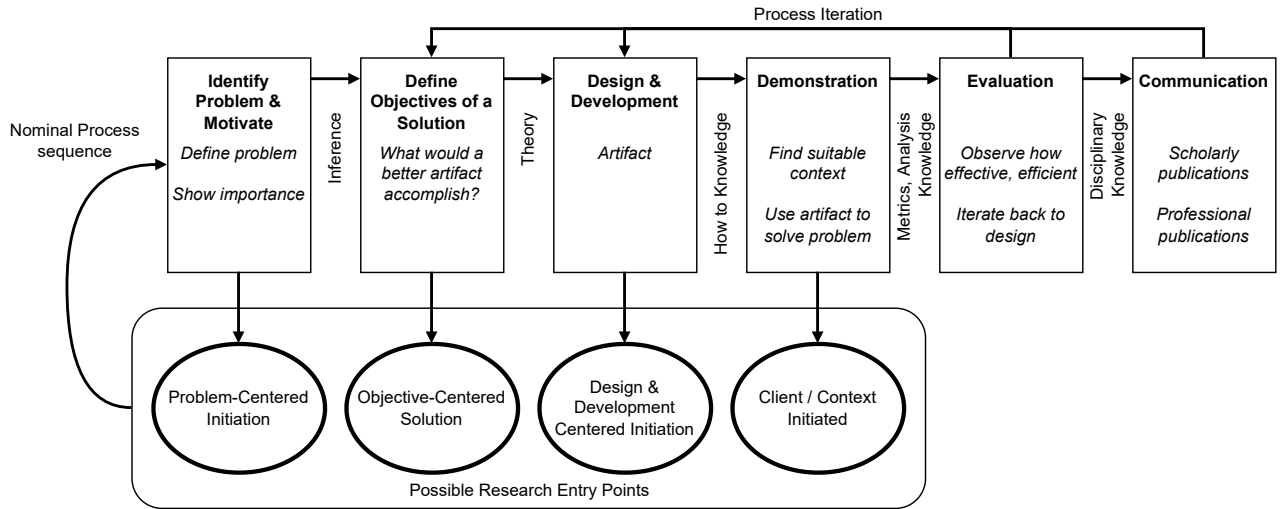


Figure 1.2: Process model of the **Design Science Research Methodology (DSRM)** [47].

The first process defines the specific research problem and justifies the value of a solution and its relevance to the scientific community. The second step defines both quantitative (based on key performance indicators and in comparison to existing solutions) and qualitative (functional description of the solution) goals from the problem definition. The following step focuses on design and development activities for creating an artifact, defining its desired functionality and architecture and then actually creating it. In the scope of this PhD thesis, an artifact can be models, algorithms, methodologies, among others. The *Demonstration* activity uses the artifact to solve one or more scenarios or case studies of the problem. Then, the next process is the observation and measurement of how well the artifact supports the solution of the problem. The *Evaluation* can be a comparison of the artifact's functionality with the objectives from the second activity in the **DSRM** model, simulations, or feedback from the end-user of the solution. Lastly, the *Communication* step focuses on the scientific dissemination of the problem and its importance, the artifact, its novelty and utility, the methodology for its design, and its effectiveness to researchers and other relevant audiences such as specialized operators. Although Peffers *et al.* [47] structures the **DSRM** model in a nominally sequential order, the methodology does not obligate the researchers to always proceed sequentially from the first to the sixth activity. Indeed, Peffers *et al.* [47] defines possible research entry points for the researchers to begin and then move outward.

1.5 Document organization

The remaining of this document is organized as follows. Chapter 2 presents a literature review on concepts and technologies related to the topics of **SLAM** and long-term localization and mapping in dynamic environments. Chapter 3 provides the problem statement and the main goals and scope of this research proposal. The statement describes the problem and formulates the research questions that should be answered during the execution of this research work and possible hypotheses that support potential answers to those questions. Chapter 4 describes preliminary work and developments. Finally, Chapter 5 outlines the work plan defining the tasks and milestones predicted for the research and the respective timeline. This chapter also discusses the scientific dissemination methodology that will be followed during the execution of the research plan.

Chapter 2

Literature Review

The preprint of a systematic literature review on long-term localization and mapping following the **Preferred Reporting Items for Systematic reviews and Meta-Analysis (PRISMA)** [48] guidelines is available in Appendix A of this document. The study precedes the elaboration of this thesis research plan and includes an analysis of 142 literature works for discussing and analyzing methodologies focused on appearance invariance, dynamic elements, map sparsification and multi-session techniques, and other computational concerns. Additionally, the study discusses commonly used public datasets and the characteristics of the experimental data used by the authors of the included works in the review while presenting the most used evaluation metrics. However, given the main scope of this research work on lifelong **SLAM** in dynamic environments, this chapter focus mainly on the discussion of localization and mapping algorithms in scenes with dynamic elements presented in the systematic review. This chapter also formulates the **SLAM** problem and its most common solutions.

Hence, this chapter is divided into four sections. Section 2.1 summarizes the methodology of the systematic review (further detail available in Appendix A) on long-term localization and mapping and overviews the included literature works in terms of publication year with an emphasis on **SLAM** in dynamic environments. Section 2.2 formulates the traditional **SLAM** problem and presents its three most common solutions: map matching, particle filter, and pose graph implementations. Then, Section 2.3 discusses literature works related to long-term **SLAM** in dynamic environments based on the findings presented in Appendix A.

2.1 Literature overview

2.1.1 Review methodology

The literature review is an important process in any research project for providing a comprehensive summary of previous research on a specific topic. This process aims to identify the main contributions, research gaps, define current challenges and future directions in the scope of a research topic. However, the review methodology should be supported on relevant and rigorous frameworks for synthesizing the findings of the literature works.

A systematic literature review uses explicit, rigorous, and reproducible systematic methods to synthesize the findings of works related to a particular research question, topic area, or phenomenon of interest. This type of review assures the quality and trustworthiness of the review’s findings by presenting a complete, organized, and summarized analysis of all works considered while allowing other researchers to replicate or update the review. The most common standard for performing systematic reviews is the **PRISMA** [48] statement. Even though the

PRISMA [48] guidelines have been designed originally for evaluating the effects of health interventions, the checklist items of the methodology are general and applicable to other subject areas. Therefore, the literature covered in this chapter and extensively discussed in Appendix A follow a systematic methodology based on the **PRISMA** [48] guidelines. This methodology is composed by the following steps:

1. definition of eligibility criteria;
2. conceptualization of the search strategy;
3. selection process:
 - (a) identification;
 - (b) screening;
 - (c) quality assessment.
4. data extraction;
5. synthesis and analysis of the included works.

The first step defines the eligibility criteria to determine the eligible studies for the selection process. In the systematic review on long-term localization and mapping, the criteria considered are mostly related with the type of paper – indexation in scientific venues, subject area, reject short papers and gray, secondary, and tertiary literature, exclude dataset papers – and availability – full-text accessibility and language. Additionally, a coverage (exclude odometry-only approaches) and scope (not consider different or non related subjects) criteria are exclusion criteria focused on the scientific contributions of the works. Next, the search strategy conceptualization defines as data sources the databases ACM Digital Library [49], Dimensions [50], IEEE Xplore [51], INSPEC [52], Scopus [53], and Web of Science [54]. The date of the last full inquiry in the systematic review was May 17, 2022. However, for the discussion of long-term **SLAM** in dynamic environment in Section 2.3, recent works between the previous data until October 25, 2022, will be included for the analysis. As for the base string used to inquiring the data sources considering the title, abstract, and keywords search fields is the following one: `(robot* OR vehicle*) AND ((locali* AND map*) OR "slam") AND ("long term" OR "life long" OR lifelong)`. The selection process starts by identifying the records in the different sources using the previous base string, possibly rejecting duplicate records (Parsifal [55] was the tool used for identifying duplicates) and also based on the eligibility criteria defined previously. The screening phase also excludes works based on the eligibility criteria by reading the title and abstract of the publications. Then, the quality assessment establishes another set of criteria focused on the scientific contribution of the eligible records. The assessment of eligible records rejects works based on works with no compliance in certain criteria (*e.g.*, non detailed methodology or results sections, and works not having long-term experiments or not considering localization nor mapping problems). Finally, the data extraction process allows the retrieval of information for the synthesis and analysis of the included works while also rejecting articles with extended versions to improve the discussion and singularity of the methodologies proposed in the literature. Figure 6 illustrates the **PRISMA** [48] flow diagram for summarizing the selection process. Further details on the systematic methodology followed for synthesizing and analysis literature works on long-term localization and mapping can be found in the Section 3 *Methodology* of the Appendix A.

2.1.2 Results overview

The relevance of the long-term localization and mapping topic can be evaluated by the evolution of the number of publications. Figure 2.2 presents this evolution using the Title, Abstract, and Keywords (authors and indexed

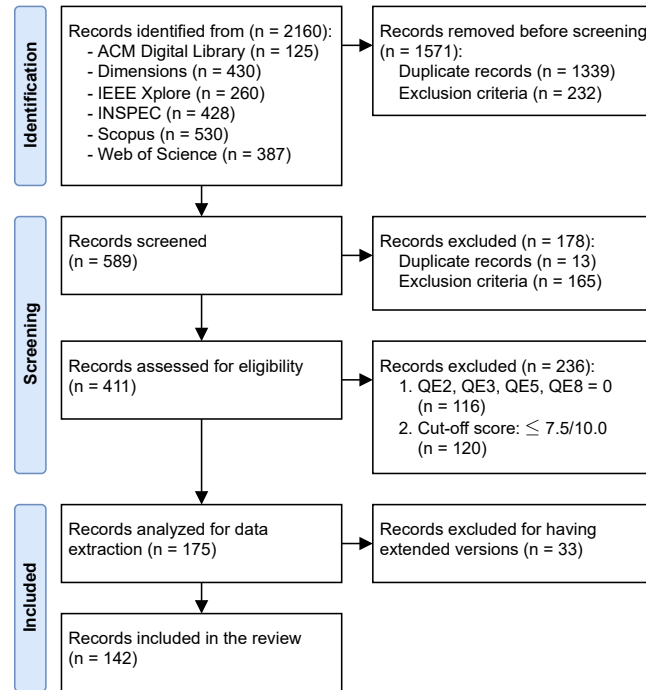


Figure 2.1: Preferred Reporting Items for Systematic reviews and Meta-Analysis (PRISMA) [48] flow diagram for the selection process of the systematic review on long-term localization and mapping. Date of the last full inquiry: May 17, 2022.

ones) as search fields when inquiring in three different bibliographic databases: INSPEC [52], Scopus [53], and Web of Science [54]. Even though the inquiry only represents the identification phase of the selection process in a systematic review (*i.e.*, applying the search strategy to all data sources), still, the search results allow having a perception of published works related to the long-term **SLAM** and the interest of the scientific community in the topic. In Figure 2.2a, the search string used for the inquiry is the same one used in the systematic review: `(robot* OR vehicle*) AND ((locali* AND map*) OR "slam") AND ("long term" OR "life long" OR lifelong)`. This string does not focus on any particular challenge related to long-term localization and mapping. In contrast, Figure 2.2b presents the evolution of the number of publications reducing the scope of the search results to the challenge of localization and mapping in dynamic environments, the same one directly related to the goals of this research proposal. This scope reduction is accomplished by using the following search string: `(robot* OR vehicle*) AND ((locali* AND map*) OR "slam") AND ("long term" OR "life long" OR lifelong) AND (dynamic* OR mov*)`. The part of the query added to the original one (`dynamic* OR mov*`) intends to limit the search results to works related to dealing with dynamic or moving elements in the scene.

When analyzing Figure 2.2, firstly, the graph in Figure 2.2a shows an increasing interest of the scientific community in the topic of long-term localization and mapping. Since 2009, the number of publications are predominantly more than 10 per year. From that year onwards, the graph has an almost linear tendency reaching a maximum of 74, 76, and 62 published records on the years of 2019, 2021, and 2021 respective to the databases INSPEC [52], Scopus [53], and Web of Science [54], respectively. The tendency shows that long-term **SLAM** is gaining relevance throughout the years, supporting the importance and relevance of the topic for the scientific community. This result is in conformity to the analysis of the year of publication presented in the systematic review for the 142 literature works included in the paper (see Section 4 *Results Overview* of the Appendix A).

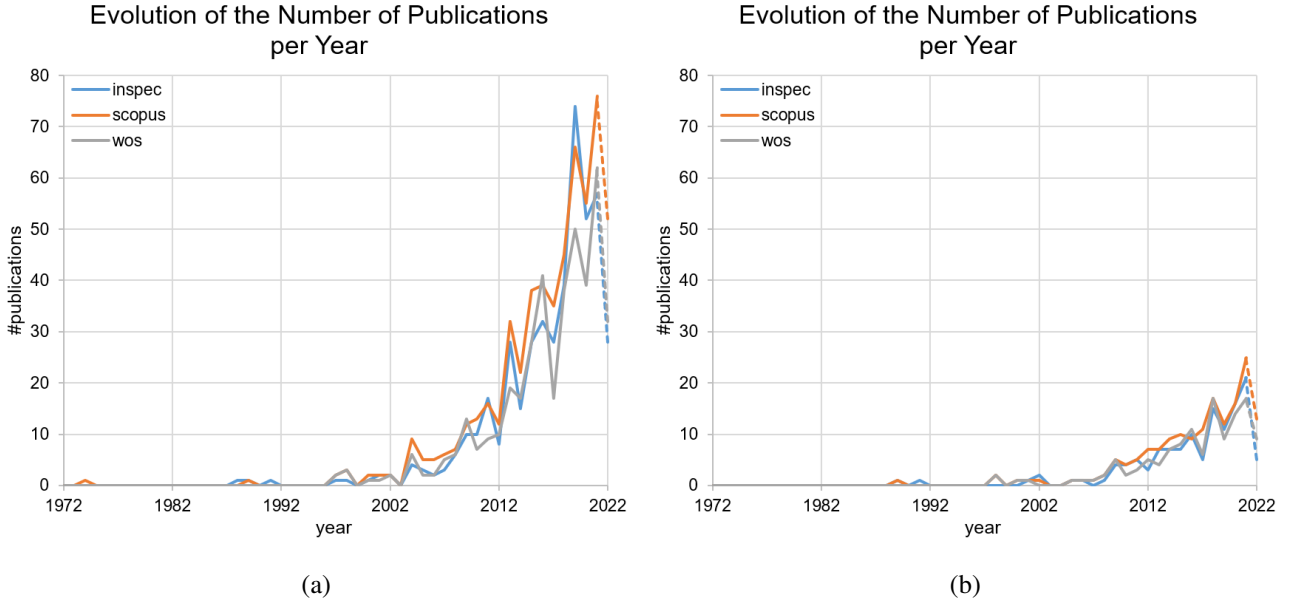


Figure 2.2: Evolution of published records per year in the INSPEC [52], Scopus [53], and Web of Science [54] databases on long-term localization and mapping: (a) (robot* OR vehicle*) AND ((locali* AND map*) OR "slam") AND ("long term" OR "life long" OR lifelong); (b) (robot* OR vehicle*) AND ((locali* AND map*) OR "slam") AND ("long term" OR "life long" OR lifelong) AND (dynamic* OR mov*). Date of the last full inquiry: October 25, 2022.

Similarly, the graph shown in Figure 2.2b indicates a growing interest in the topic of this research proposal, *i.e.*, long-term localization and mapping in dynamic environments. The topic also seems to have gained relevance from 2009 onwards, achieving a maximum of 21, 25, and 17 published works in 2021 respective to the databases INSPEC [52], Scopus [53], and Web of Science [54], respectively. Although the number of publications related to dynamic environments (Figure 2.2b) is lower than the ones associated with long-term SLAM (Figure 2.2a), the growing interest in solving the challenges of performing localization and mapping in dynamic environments demonstrates the relevance of this research proposal.

2.2 Simultaneous Localization and Mapping (SLAM)

Next, this section focuses on defining the SLAM problem and discussing the three main techniques for SLAM: map matching, particle filter, and pose graph. Given the focus of this thesis research plan on maintaining an up-to-date representation of the environment when operating AMRs in dynamic surroundings, the algorithms to be developed should not only localize the robot in the current map but also update the latter accordingly to the dynamics of the scene elements. Thus, the problem of long-term localization and mapping in dynamic environments is intrinsically related to SLAM.

2.2.1 Problem introduction

SLAM is one of the most fundamental problems in robotics. This problem, also previously known as **Concurrent Mapping and Localization (CML)** in the literature, may be formulated by the Definition 2.2.1. When the robot does not have access to a representation of the environment nor know its pose, SLAM enables an AMR to

build a map of its surroundings while simultaneously localizing itself relative to this map. **SLAM** may also be utilized for building the map used in applications assuming a static scene, even if the map is only continuously updated upon the mapping process [56].

Definition 2.2.1 ([3, p. 349]). “...the robot must not only create a map, but it must also do this while moving and localizing to explore the environment. In the robotics community, this is often called the *Simultaneous Localization and Mapping (SLAM)* problem.”

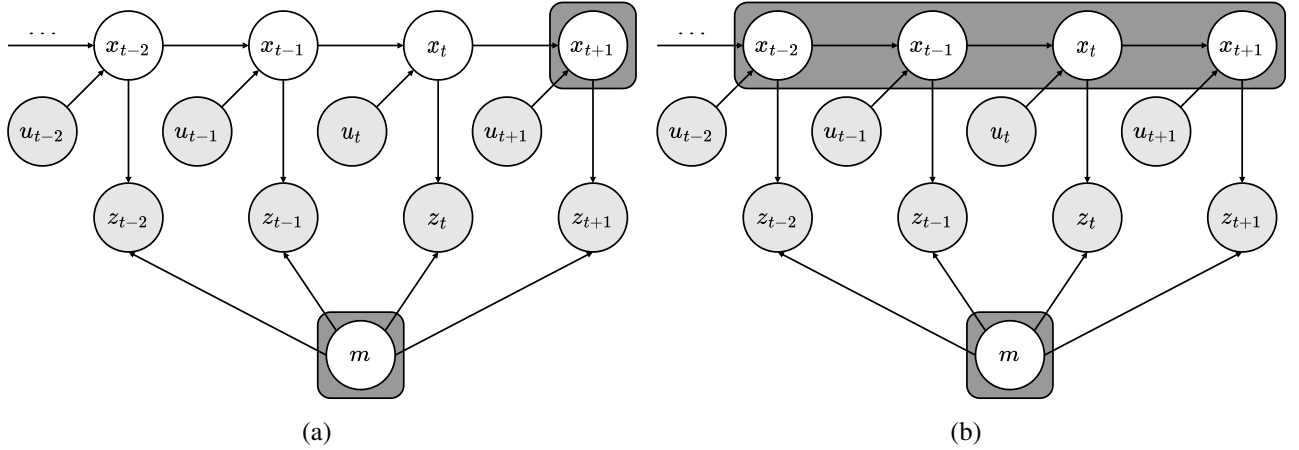


Figure 2.3: Graphical model of the **Simultaneous Localization and Mapping (SLAM)** problem: (a) online SLAM; (b) full SLAM [56].

Two main formulations define **SLAM** from a probabilistic perspective, also illustrated graphically in Figure 2.3. The first represented in Figure 2.3a is online **SLAM**, where the algorithm estimates the posterior over the momentary pose (x_t at a time instant t) along with the map (m), as defined in Equation 2.1, provided the sensor measurements ($z_{1:t}$) and the control variables ($u_{1:t}$). The online term is due to only involving the estimation of variables that persist at the current time instant t . Several algorithms incrementally implement online **SLAM**, *i.e.*, discarding past measurements and controls once they have been processed. The other probabilistic formulation is called full **SLAM** and illustrated in Figure 2.3b, given the goal to compute a posterior over the entire path ($x_{1:t}$) along with the map, as formulated in Equation 2.2, instead of just the current pose (x_t). The difference between online and full **SLAM** is that the former results in an integration of the past poses from the full **SLAM** problem [56].

$$p(x_t, m \mid z_{1:t}, u_{1:t}) \quad (2.1)$$

$$p(x_{1:t}, m \mid z_{1:t}, u_{1:t}) \quad (2.2)$$

In the case of the algorithm also estimating correspondences between measurements and map landmarks, the explicit inclusion of the correspondence variables (c_t or $c_{1:t}$) in the estimation problem may be useful for reasoning if the perceived feature is the same one as previously detected and already represented in the map or if it is a new landmark. Consequently, Equations 2.3 and 2.4 formulate the posterior estimation for online and full **SLAM**, respectively [56].

$$p(x_t, m, c_t \mid z_{1:t}, u_{1:t}) \quad (2.3)$$

$$p(x_{1:t}, m, c_{1:t} \mid z_{1:t}, u_{1:t}) \quad (2.4)$$

Although computing a full posterior captures all the information to be known about the map and the pose or the path, this estimation is practically unfeasible due to the high dimensionality of the continuous parameter space and the large number of discrete correspondence variables [56]. So, next, the discussion on scan matching, particle filter, and pose graph implementations of **SLAM** focuses on the practical aspects of performing continuous localization and mapping.

2.2.2 Scan matching

A basic approach for solving the **SLAM** problem is continuously matching the sensor scans to the map learned until the current instant. An example is the HectorSLAM [57] algorithm, where the block diagram describing that system is shown in Figure 2.4. This algorithm is considered to be online **SLAM** due to only estimating the current pose and the map. Also, the method integrates incrementally the sensor readings to allow discarding those in subsequent iterations. Furthermore, HectorSLAM [57] aligns the beam endpoints of a 2D **LiDAR** with the current map using a matcher, *e.g.*, PerfectMatch [58], **Iterative Closest Point (ICP)** [59], **Normal Iterative Closest Point (NICP)** [60], **Normal Distribution Transform (NDT)** [61], or other matchers. This alignment estimates the homogeneous transformation between the sensor scan and the map for updating the map with the latest scan and also update the current estimation of the robot pose.

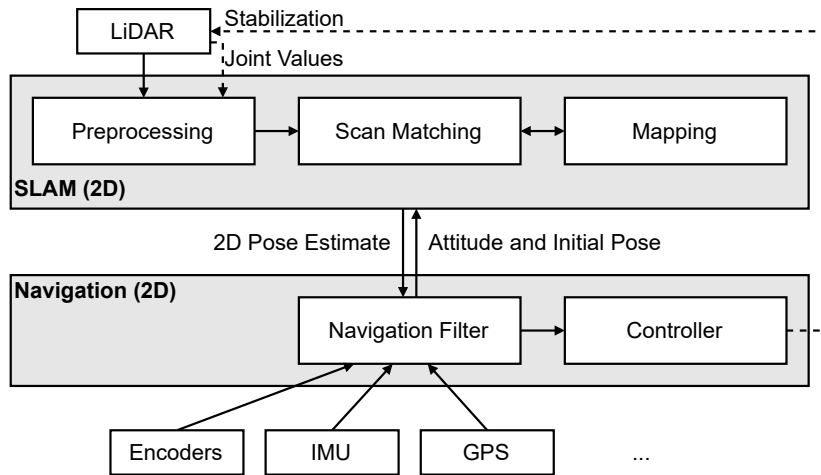


Figure 2.4: Block diagram of the mapping and navigation systems of Hector **SLAM**. Dashed lines represent optional information [57].

The main advantage of scan matching **SLAM** algorithms is the lower use of computational resources compared to more complex approaches. These algorithms do not require an exhaustive pose search or data association search with previous scans (the matching with all preceding scans is implicit in matching the current one

with the map) [57]. However, as in the case of HectorSLAM [57], a pure scan matching approach does not implement loop closure (see Definition 2.2.2). This limitation does not allow correcting the map when recognizing a previously mapped place, not having the consequent reduction in the uncertainty of the map estimate.

Definition 2.2.2 ([3, p. 188]). “The problem of location recognition is also called loop-detection, because a loop is a closed trajectory of a vehicle that returns to a previously-visited point. The problem of minimizing the error at the loop closure is instead called loop-closing.”

2.2.3 Particle filter

The graphical model of the SLAM problem shown in Figure 2.3 is equivalent to a discrete Dynamic Bayesian Network (DBN) (see Definition 2.2.3). A sampling-based inference algorithm for DBNs is the particle filter. However, these filters suffer from the problem of dimensionality scaling exponentially [62]. The assumption of conditional independence between map measurements allows using Rao-Blackwellized Particle Filters (RBPFs) for SLAM. These filters use particles to represent the posterior over some variables and Gaussians (or other parametric probabilistic distribution functions) to represent all other variables [56]. This marginalization of some variables implemented by RBPFs improves the computational efficiency compared to standard particle filters, reducing the space size for the sampling step [62].

Definition 2.2.3 ([63]). “A convenient way to describe this structure is via the Dynamic Bayesian Network (DBN)... A Bayesian network is a graphical model that describes a stochastic process as a directed graph. The graph has one node for each random variable in the process, and a directed edge (or arrow) between two nodes models a conditional dependence between them”

An example of a RBPF-based SLAM is FastSLAM [64, 65]. This algorithm decomposes the SLAM problem into a robot localization one and the estimation of map landmarks conditioned on the robot pose estimate. Figure 2.5 represents the composition of particles implemented in FastSLAM. Each particle k contains an estimate of the robot path ($x_{1:t}^{[k]}$) until the current time instant t and a set of Extended Kalman Filters (EKFs) with a mean ($\mu_j^{[k]}$) and covariance ($\Sigma_j^{[k]}$) of the estimation, one for each feature $j = 1..N$ represented in the map m . The basic steps of the FastSLAM [64, 65] algorithm are the following ones [56]:

- for each particle $k = 1..M$:
 1. **retrieval**: retrieve a pose $x_{t-1}^{[k]}$ from the particle set;
 2. **prediction**: sample a new pose $x_t^{[k]} \sim p(x_t | x_{t-1}^{[k]}, u_t)$ using the probabilistic motion model of the robot (the improved version, FastSLAM 2.0 [65], incorporates also the sensor measurements z_t^i in this step);
 3. **measurement update**: for each observed feature obtained from a measurement z_t^i , identify the correspondence j in the particle map, and incorporate z_t^i into the corresponding EKF, updating the mean ($\mu_j^{[k]}$) and covariance ($\Sigma_j^{[k]}$) of the corresponding map feature accordingly;
 4. **importance weight**: compute the importance weight $w^{[k]}$ for the new particle.
- **resampling**: sample, with replacement, the M particles, where the probability of sampling being proportional to $w^{[k]}$, to incorporate the measurements z_t^i into the robot pose $x_t^{[k]}$ (due to only considering the most recent control u_t in the previous steps, only in FastSLAM 1.0 [64]).

The adaptation of FastSLAM [64, 65] to occupancy grid-based SLAM is implemented by the GMapping [66] algorithm. This adaptation requires the modification of three modules relative to FastSLAM [64, 65].

	robot path	feature 1	feature 2	...	feature N
Particle $k = 1$	$x_{1:t}^{[1]} = \{(x \ y \ \theta)^T\}_{1:t}^{[1]}$	$\mu_1^{[1]}, \Sigma_1^{[1]}$	$\mu_2^{[1]}, \Sigma_2^{[1]}$	$\dots$	$\mu_N^{[1]}, \Sigma_N^{[1]}$
Particle $k = 2$	$x_{1:t}^{[2]} = \{(x \ y \ \theta)^T\}_{1:t}^{[2]}$	$\mu_1^{[2]}, \Sigma_1^{[2]}$	$\mu_2^{[2]}, \Sigma_2^{[2]}$	$\dots$	$\mu_N^{[2]}, \Sigma_N^{[2]}$
		$\vdots$			
Particle $k = M$	$x_{1:t}^{[M]} = \{(x \ y \ \theta)^T\}_{1:t}^{[M]}$	$\mu_1^{[M]}, \Sigma_1^{[M]}$	$\mu_2^{[M]}, \Sigma_2^{[M]}$	$\dots$	$\mu_N^{[M]}, \Sigma_N^{[M]}$

Figure 2.5: Composition of particles in FastSLAM [64, 65] in terms of path estimate and the set of estimators and associated covariances for individual feature locations [56].

First, GMapping [66] considers the occupancy grid as the map representation for the **SLAM** problem, where each particle contains a grid map. Next, the computation of a measurement likelihood ($p(z_t | x_t^{[k]}, m_{t-1}^{[k]})$) uses the beam endpoint sensor model, characteristic of range-bearing sensors such as 2D lasers, instead of integrating the measures into an **EKF** as in FastSLAM [64, 65]. This model considers the individual scan beams independent from each other, using the distance between the beam endpoint and the closest obstacle in the map to that point to compute the beam's individual likelihood. Lastly, the update of each particle's map ($p(m_t^{[k]} | z_t, x_t^{[k]}, m_{t-1}^{[k]})$) computes a new occupancy grid map. This update estimates the probability of each individual cell, assuming the independence of each other, using a binary Bayes filter and the inverse model of the sensor measurements [56].

A key advantage of particle filter-based **SLAM** is the decisions on data associations being made on a per-particle basis, instead of the most likely one as in scan matching or **EKF** approaches. This characteristic represents inherently multi-hypotheses associations in the filter, making it more robust to data association problems. Another advantage is particle filters being capable to model non-linear robot motions without requiring model approximations. However, a disadvantage of particle filters is relative to loop closure detection. Even though these filters are able to detect previously visited places, unlike scan matching approaches such as HectorSLAM [57], this place recognition ability is only maintained through the diversity in the particle sets, possibly maintaining larger sets and requiring more computational resources [56].

2.2.4 Pose graph

Another solution for the **SLAM** problem leveraging its graphical model (illustrated in Figure 2.3) is pose graph **SLAM**, also known as GraphSLAM. This optimization-based solution consists of two subsystems, as in filter-based approaches such as particle filters and **EKF**. In filter-based **SLAM**, the filters implement a prediction stage usually based on the vehicle's motion model, and then, the correction phase updates the predicted state by matching the sensor observation to the map. As for optimization-based solutions such as the pose graph formulation, a first module also known as the front-end is responsible to identify constraints of the problem based on sensor data by finding correspondences between new observations and the current map. Then, a second subsystem known as the back-end computes or refines the poses of the graph and the map minimizing

the coherence errors as a whole. This second module usually requires optimization algorithms such as Gauss-Newton or Levenberg-Marquardt for minimizing the objective function [67]. However, there are available in the literature optimization more specific for the pose graph formulation, *e.g.*, **Tree-based netwORk Optimizer (TORO)** [68], **incremental Smoothing And Mapping (iSAM)** [69, 70], or g^2o [71].

Furthermore, the edge connection in GraphSLAM between vertexes x_i and x_j is illustrated in Figure 2.6. Considering z_{ij} as a virtual measurement obtained from raw sensor data between those two vertexes, let z_{ij} and Ω_{ij} be the mean and the information matrix of the virtual data, respectively. This virtual measurement actually represents the transformation that maximizes the overlap between the observations acquired from the pose nodes i and j . The log-likelihood l_{ij} (see Definition 2.2.4) of the measurement z_{ij} can be defined as in Equation 2.5, where $\hat{z}_{ij}(x_i, x_j)$ is the prediction of the virtual measurement given the configuration of the nodes x_i and x_j . Then, the configuration of nodes $x^* = (x_1, \dots, x_T)^T$ can be found formulating a maximum likelihood problem that minimizes the negative log-likelihood function $F(x)$ (see Equation 2.6) of all the observations, seeking to solve the Equation 2.7 [63]. For exemplification purposes, two possible examples of virtual measurements are wheeled odometry and 2D laser scan matching data. Both cases would define the prediction of the measurement as $\hat{z}_{ij} = x_j - x_i$. The virtual measurement would be $z_{ij} = x_j - g(x_i, u_{ij})$ for wheeled odometry, where $g(x_i, u_{ij})$ represents the odometry estimation of x_j starting from x_i , while using the wheeled odometry data u_{ij} between those two nodes. As for the scan matching case, the measurement would be $z_{ij} = h(z_i, z_j)$ representing the matching between the laser scans z_i and z_j acquired from the nodes x_i and x_j , respectively.

Definition 2.2.4 ([72, p. 224]). “...since the natural logarithm is a one-to-one function on $\mathbb{R}^+$, no information about θ is lost when the log-likelihood function $l(\theta) = \ln[p(\theta)]$ is used in place of $p(\theta)$. Moreover, for many probability models, it is easier to work with the log-likelihood function than with the likelihood function...”

$$l_{ij} \propto [z_{ij} - \hat{z}_{ij}(x_i, x_j)]^T \Omega_{ij} [z_{ij} - \hat{z}_{ij}(x_i, x_j)] \quad (2.5)$$

$$F(x) = \sum_{i,j} e_{ij}^T \Omega_{ij} e_{ij} \quad (2.6)$$

$$x^* = \underset{x}{\operatorname{argmin}} F(x) \quad (2.7)$$

A pose graph **SLAM** implementation is the **SLAM** Toolbox framework [73], also available in the official navigation stack of the **Robot Operating System (ROS)** [74, 75] (version ROS2). This toolbox is intended for 2D **SLAM** using a 2D laser scanner in conjunction with wheeled odometry obtained from the wheel encoders of the robot’s motors. The optimization framework used for the pose graph back-end is the Ceres Solver [76]. Although **SLAM** Toolbox [73] states having lifelong mapping capabilities, these are only relative to being able to update existing maps and serialize the data for use in other mapping sessions, not for dynamic environments or appearance invariance.

Overall, a key difference between pose graph **SLAM** compared to other solutions for the **SLAM** problem, such as filter and scan matching approaches, is that the former has access to the full data when building the map, being able to apply improved linearization and data association techniques. Indeed, in GraphSLAM, all data can be used to linearize and calculate correspondences, *i.e.*, can revise past data association and linearize more than once [56]. This key difference is pointed as a possible reason to graph **SLAM** tending to give better results than

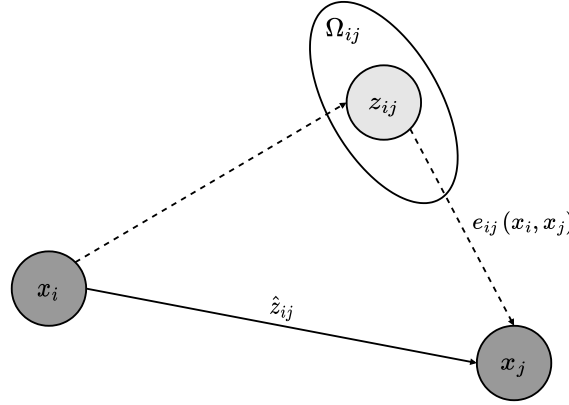


Figure 2.6: Edge connection in the graph formulation of the **SLAM** problem between vertexes x_i and x_j originated from a measurement z_{ij} [63].

filters, where the latter is more subject to linearization issues [67]. However, the pose graph formulation shows a linear growth over time in the graph's size requiring techniques for constraining the graph size (see Section 1.1), and thus, limit the computational resources required for continuously performing GraphSLAM [56].

2.3 Long-term SLAM in dynamic environments

In terms of works in the literature for long-term **SLAM**, as previously mentioned, Appendix A includes a systematic literature review on that topic following the **PRISMA** [48] guidelines. The literature review goes over several challenges related to long-term autonomy having the date of the last full inquiry for identifying records in the data sources considered in the study on May 17, 2022. Given the focus of this research proposal on the challenge for performing **SLAM** in dynamic environments, this section focus on the included works of the review study categorized as dynamic in the data extraction phase of the methodology. Furthermore, another full inquiry was performed on October 25, 2022, to include more recent works published or made available online between the date considered in the review up to the latter one. As a result, after executing all stages of the review methodology described in Appendix A and focusing only on dynamic-related works, another five records were found in the literature ([77–81]). However, given that the work [80] is an extension of [82], as assumed in the review, only the extended version will be considered for the discussion.

So, next, this section analyzes literature works focused on modeling and identifying dynamic elements in the environment within the scope of long-term **SLAM**, *i.e.*, how the methods identify these elements and handle them for long-term localization and mapping. The discussion on dynamics modeling is organized into the following topics: specific map representations used to model or deal with dynamic elements in the scene, identification of dynamic elements matching the current observation to the current map, future prediction of dynamic properties of scene elements, and semantic identification of dynamic objects.

2.3.1 Map representation

Inspired by the human memory, Dayoub *et. al.* [30] and Bacca *et. al.* [31] adapt the multi-store model of Atkinson and Shiffrin [36] for robot mapping. This model divides the memory into three stores: **Sensory Memory**

(SM) to save the perceived information, **STM**, and **LTM**. Three mechanisms move information between memories: selective attention for **SM** to **STM**, rehearsal to commit information from **STM** to **LTM** or which one is forgotten, and the retrieval mechanism to move unused information from **LTM** back to **STM**. Dayoub *et. al.* [30] implements two types of state machines for rehearsal and retrieval mechanisms of the **STM** and **LTM**. In rehearsal, a **STM** feature moves closer to **LTM** or moves back to the initial state (or forgotten if already in that state) when observed consecutively or if not, respectively. Similar for retrieval, where a feature in **LTM** moves to the initial state or closer to forget if observed in the current view or not. Consequently, **LTM** and **STM** save the most static and dynamic features based on their observability in the current view, respectively. In a changing environment, the method decreased the localization failure rate compared to a static view. Instead of using a state machine, Bacca *et. al.* [31] implements a **Feature Stability Histogram (FSH)** depending on the feature observability to distinguish between **STM** and **LTM** features using k -means clustering. This modification allows that an input feature in **SM** can bypass **STM** and become part of **LTM** depending on the feature strength. The method was able to filter out pedestrians from 2D laser and camera data and achieved a more accurate representation of the environment compared to a static approach.

Although Biber and Duckett [5] does not adopt specific memory mechanisms, they implement **STM** and **LTM** maps. The method implements a dynamic map as a set of local maps, each maintaining submaps representing different timescales. The timescale parameter of each submap determines probabilistically when to add samples from 2D laser scans. The dynamics of the environment are represented by using 5 different timescales, where the smaller one ($\sim 3.1s$) represents an **STM** map updated at every instant and the other 4 timescales ($\sim 0.43runs$, $0.43days$, $3.1days$, and $13.5day$) are **LTM** submaps updated after each session or daily. Instead of only localizing on the **LTM** maps as Dayoub *et. al.* [30] and Bacca *et. al.* [31], Biber and Duckett [5] selects the best representation of both **STM** and **LTM** maps that best explain the sensor data. In a 5 weeks experiment, the localization with a dynamic map improved while a static representation degraded over time, while the timescales led to static parts as walls emerging in **LTM** and dynamic elements disappearing from **STM** maps.

Similar to the works [30] and [31], two maps can represent a more stable and a more dynamic representations of the environment. In Walcott-Bryant *et. al.* [24], an active map represents the most current state of the environment, including parts that did not change from previous passes and objects added to the environment. A dynamic map only saves the points of a 2D laser scan that changed over time. Wang *et. al.* [34] uses a tracking map with short-term static points and a long-term map only containing long-term static points identified by a semantic segmentation module with ORB-SLAM2 [83]. Also, Zhu *et. al.* [35] creates offline a semi-dynamic map and a static one, where the former has semi-dynamic objects (parked cars in a parking lot environment) and the latter has both static and semi-dynamic objects. The main goal of representing two different dynamics is usually to favor the most stable one in the long-term. Walcott-Bryant *et. al.* [24] and Wang *et. al.* [34] only use static parts in the most current representation of the environment (active and the long-term maps, respectively) for localization. In the experiments, the approach [24] showed that their method was able to identify static parts, although it was affected by false positives and negatives, and by the blur effect in the 2D grid map. The method [34] improved the **Absolute Trajectory Error (ATE)** in a dynamic environment over ORB-SLAM2 [83] and DynaSLAM [84]. As for Zhu *et. al.* [35], its **Monte Carlo Localization (MCL)** framework (equivalent to particle filter-based localization) reduces the weight corresponding to observations of moved semi-dynamic objects. The method improved the localization in a parking lot compared to a standard **MCL**.

As for a multi-hypothesis representation, Nielsen and Hendebay [78] propose a framework for tracking the landmarks positions using Markov chains to model the transition between positions. The method [78] groups

the positions hypotheses based on the landmark's signature, where the position with the highest probability within a group according to incoming observations is selected as the active one. The hypothesis likelihood depends on the matched measurements to the landmark in the map and also on the number of missed detections, *i.e.*, landmarks with a probability of being seen in the current pose but not represented in the observations. Although the method [78] maintained an up-to-date representation of the actual landmark positions in simulated experiments, the experiments assumed always uniquely identifiable landmarks.

2.3.2 Map matching

Environment dynamics can be identified by comparing the current observation to the map. Assuming a prior vector map as a permanent map, the work of Biswas and Veloso [85] determines the probability of observed features being long-term ones by the 2D laser scan-to-map matching distance. Short-term features are determined by the scan-to-scan matching distance, while the remaining ones are considered dynamic features and not considered for localization. Compared to **MCL** with a static map and to Tipaldi *et al.* [41], Biswas and Veloso [85] obtained lower localization error in a parking-lot environment. However, the method [85] would not handle semi-static changes. Instead of using a permanent map, Zhang *et al.* [86] maintains a **Signed Distance Field (SDF)** representation based on a prior occupancy map. The proposed approach [86] rejects dynamic points identified by range flow and updates the **SDF**-based map with semi-static changes observed in the scan-to-map difference. Compared to **MCL** in a semi-static environment, Zhang *et al.* [86] had lower pose errors and an improved representation of the environment. Similarly, Fu *et al.* [81] uses **SDFs** in the mapping process, proposing the **PlaneSDF** representation where a scene is a set of planes and each one has associated a **SDF** volume. This volume describes the geometric details of the objects attached to its associated plane. Then, a change detection algorithm performs a 2D height map comparison and a 3D geometric validation between two **PlaneSDF**-based maps to compute a change mask. However, the method [81] failed to detect changes in ambiguous situations such as object occlusions in the height comparison direction. Boniardi *et al.* [25] detects semi-static changes leveraging the **ICP** scan-to-map consistency and a **Computer-Aided Design (CAD)** prior of the environment, and updates the map accordingly. The method [25] was capable of maintaining a consistent map when dealing with substantial reconfiguration of the environment. Du *et al.* [87] minimizes the Gibbs energy defined on the proposed **Long-term Consistent Conditional Random Field (LC-CRF)** for detecting dynamic points, considering that these points have often a large reprojection error in frame-to-map matching and points tend to have the same dynamic properties as the neighbor ones. In a dynamic scene, **LC-CRF** [87] achieved lower **ATE** than ORB-SLAM [88]. Unlike **LC-CRF** [87], the **Dynamic visual Feature Removal Simultaneous Localization and Mapping (DFR-SLAM)** proposed by Liu *et al.* [77] does not look for similar dynamic properties in the features neighbor. Instead, the method [77] assumes that static features have consistent motion over long time and dynamic ones have different motion patterns, labeling them using the proposed motion consensus filtering algorithm. **DFR-SLAM** [77] formulates binary classification problem based on a graph-cut optimization framework, assigning penalties to connected visual features between different instants that do not share the same label. The method [77] had lower **ATE** than ORB-SLAM2 [83] in dynamic scenarios.

Furthermore, Pan *et al.* [89] and Ding *et al.* [90] leverage clustering properties of the observations evaluating the observations count. The approach [89] segments the points of a 3D **LiDAR** point cloud into different clusters assuming that dynamic points do not appear frequently in the same place. The map only considers clusters that appear in same location more than 10 times. As for Ding *et al.* [90], the method builds on the

assumption that dynamic and static parts of the environment have a clustering property relative to its neighbors (similar to Du *et. al.* [87]). The number of observations in different sessions combined with its consistency relative to its neighbors determine if a map point is static throughout the sessions. Both representations of the environment in Pan *et. al.* [89] and Ding *et. al.* [90] were stable to structural changes in the environment.

The concept of ray tracing is also used by the included works to handle dynamic changes. Lázaro *et. al.* [91] uses ray tracing to exploit the free space information. When comparing two 2D point clouds from a viewpoint, the ray tracing evaluation identifies new objects added to the scene (observed point closer to viewpoint than the old one) and outdated information (observed point further away), allowing the identification of dynamic changes and having an up-to-date representation for localization. Given that ray tracing in 3D is expensive in terms of memory and requires dense map representations, Pomerleau *et. al.* [92] uses directly the sparse point cloud from a 3D laser. The map points are associated with each single reading in small conical apertures in spherical coordinates, updating the observed points closer than the mapped ones and the further ones are left untouched. The approximation of ray tracing results are used to update the probability of points in the map to be dynamic, based on a Bayesian approach. The probability of being dynamic can be used in ICP to not trust dynamic points and indeed, in the experiments, the work [92] had a more precise and cleaner map of the environment than using a standard ICP matching. Instead of weighting the map points, An *et. al.* [93] proposes the **Dynamic Edge Link (DEL)** to model the dynamics in the edges of a pose graph instead on the data itself. The observation of moving obstacles between two poses change the weight of the respective edge, decrease gradually the weight until not detecting the obstacle. Integrating DEL in an exploration scheme, nodes with a edge weight average lower than a certain threshold, meaning frequent moving obstacles or changed structure near that node, are not considered for exploration due to the robot may be unable to move to that position.

Although the standard NDT representation does not model free space, the works of Einhorn and Gross [94, 95] and Saarinen *et. al.* [96] use NDT with occupancy maps to model explicitly the free space and adopt exponential weighted moving average and covariance for new measurements having an higher influence than old ones. Einhorn and Gross [95] proposes a generic 2D/3D mapping using NDT and occupancy maps. The hit cells considering the current observation are updated incrementally with exponential weighting. The other cells along the sensor beam potentially empty are updated using the standard update rule of occupancy maps based on the log-odds of the occupancy value and on the inverse range sensor model. Instead of using the standard occupancy map update, the sensor model in Saarinen *et. al.* [96] depends on the inconsistency between observation and map. Also, the occupancy value describes the confidence of the NDT based on past observations. As for Einhorn and Gross [95], the method defines two probabilities for the occupancy map: occupancy and statically occupied, where the first is updated based on the sensor model (2D/3D generic beam sensor), and the second one is adapted slowly to high probability for static objects in the environment. The statically occupancy probability follows the proposed ad-hoc model that is parameterized to control how fast the static occupancy probabilities are adapted, depending also on the occupancy probability itself. The works [94] and [95] were able to handle semi-static and dynamic changes having a consistent and up-to-date representation of the environment, while the approach [96] favored long-term static structures in dynamic environments.

2.3.3 Dynamic objects detection

In terms of detecting dynamic objects, Yue *et. al.* [37] proposes a collaborative dynamic mapping for detecting humans using visual and thermal images and a 3D LiDAR. The YOLOv3 algorithm extracts the bounding

boxes from the images relative to humans. The 3D point cloud projection onto the images allows the creation of a static point cloud for localization and mapping of each robot by filtering out the points corresponding to humans. In the experiments, the dynamic objects removal generated a more accurate relative transformation of the collaborative maps compared to not removing those objects. Zhu *et al.* [35] also uses YOLOv3 to extract bounding boxes of dynamic classes (parked cars) from a RGB image, and the projection of 3D LiDAR points allows the creation of the static and semi-dynamic maps required for localization. Even though the method [35] does not discard the dynamic objects, the localization module reduce the importance of weight of the corresponding observations. Instead of using visual data, Sun *et al.* [97] adapts the PointNet network for object recognition (pedestrian, cyclist, car, or background) to classify the scan points of a 3D LiDAR. The proposed Recurrent-OctoMap maintains the occupancy and semantic information, where the latter information specifies the cell semantic state and the probability of the prediction. The transition between states is learned by a LTM Recurrent Neural Network (RNN). In a long-term experiment, the method [97] was able to improve its 3D semantic map compared to a standard Bayes update.

Moreover, pixel-wise semantic segmentation is another way to identify dynamic objects. Additionally to the semantic-descriptor proposed by Singh *et al.* [38] describing the spatial distribution of semantic entities in the scene, the method sets lower weights to features detected on sky and dynamic classes (person, car, *etc.*) from the semantic segmentation of images using EdgeNet. The proposed approach [38] obtained higher accuracy when compared to SeqSLAM [98], FAB-MAP [99], and Bag-of-Words (BoW) [100]-based place recognition methods in a highly dynamic outdoor experiment. Instead of identifying object classes, Song *et al.* [39] proposes the MD-Net CNN to segment a grayscale image into unstable, static, and moving pixel points, only using static points for localization. The localization error was reduced compared to not estimated the pixel dynamic attribute. Ganti and Waslander [40] proposes the Semantically Informed Visual Odometry (SIVO) to improve the performance of ORB-SLAM2 [83] by using the Bayesian neural network SegNet for segmentation and computation of the network uncertainty. SegNet is trained to distinguish different object classes for identifying dynamic objects (sky, car, truck/bus, person/rider, motorcycle/bicycle, and void) from static ones (road, traffic sign, building, wall/fence, pole, vegetation, sidewalk, traffic light, and terrain). Only static keypoints that reduce the most of the state's uncertainty (considering the network uncertainty) are considered as input for ORB-SLAM2 [83]. SIVO [40] was able to remove uninformative and dynamic keypoints from the current frame. However, the rejection of potential dynamic objects without verifying if they are moving reduced the localization performance of the module in certain scenarios.

Dynamic objects identification can be improved by verifying geometric constraints. Bescos *et al.* [84] proposes DynaSLAM, using multi-view geometry verification for improving semantic labeling of potential dynamic classes obtained from a Mask RNN. Although DynaSLAM [84] outperformed ORB-SLAM2 [83] in highly dynamic scenarios, its performance reduces in slower dynamics. Similarly, Wang *et al.* [34] also uses a geometric verification for improving labeling of dynamic objects, the latter obtained by a ResNet-based network. The verification is based on the reprojection error. The method [34] improved the ATE over DynaSLAM [84] and ORB-SLAM2 [83] in a scenario with movable objects. Guo and Liu [79] also uses a Mask RNN network for pixel-wise segmentation and object instance labels, similar to DynaSLAM [84]. Rather than using multi-view geometry verification just for improving semantic labeling, this verification is used in [79] also for detecting untrained moving objects. The approached [79] outperformed DynaSLAM [84] in terms of ATE in dynamic scenes. Instead of using semantic segmentation, the Semantic and Geometric Constraints Visual Simultaneous Localization and Mapping (SGC-VSLAM) [101] proposed by Yang *et al.* uses YOLOv3 to ex-

tract bounding boxes of dynamic objects for also improving ORB-SLAM2 [83]. A constraint based on epipolar geometry improves the labeling. **SGC-VSLAM** decrease the **Root Mean Square Error (RMSE)** of the **ATE** by 96% compared to ORB-SLAM2 [83] in highly dynamic environments. However, similar to DynaSLAM [84], its performance decreased in lower dynamics. Finally, Xing *et. al.* [102] proposes the DE-SLAM to deal with **Short-Term Dynamics (STD)** and **Long-Term Dynamics (LTD)** at the same time. A MobileNetv2 identifies bounding boxes of movable objects (cars, persons, *etc.*) classified as **STD**. A motion check of **STD** elements recognizes all moving objects in the current keyframe. As for **LTD**, DE-SLAM uses **Histogram of Oriented Gradients (HOG)** features extracted from ORB keypoints to improve its invariance to illumination changes. In the experiments, DE-SLAM [102] improved the localization over ORB-SLAM2 [83] in a changing environment. All of these methods using geometric constraints to improve semantic labeling only use static features for localization and mapping.

2.3.4 Prediction modeling

In the included works, Markov processes are used to predict the dynamics of the environment. Tipaldi *et. al.* [41] uses a dynamic occupancy grid and exploits the stationary distribution and the state holding time associated with **Hidden Markov Models (HMMs)** on a 2D grid. The method uses past observations for each run to learn the state transition probabilities iteratively to estimate the **HMM** parameters. Then, the localization can infer how often is expected to see a dynamic object in the environment and for how long. Comparing the proposed **HMM**-based localization to **MCL** using a standard grid, the former had a lower localization failure rate than **MCL**, capable of dealing with high dynamics (moving cars) and lower ones (parked cars). Rapp *et. al.* [42] implements a semi-Markov process extended by a Levy process to model a time dependency on the state holding time of Markov processes, also predicting as Tipaldi *et. al.* [41] the expected retention time for each cell being in a specific state. In the experiments, the proposed model [42] integrated in **MCL** improved the classic **MCL** in a dynamic environment.

The environment dynamics can have periodic patterns associated with them. Assuming periodic changing patterns, Krajník *et. al.* [43] proposes **FreMEn** to model the probability of occupancy or feature visibility in a grid as a combination of harmonic functions related to periodic processes. **FreMEn** [43] uses spectral analysis (Fourier transform) to compute the harmonic functions and predict future state with a given confidence. In a changing environment, **FreMEn** [43] outperformed a static map in terms of localization error by selecting the most likely visible features at each location for localization. Santos *et. al.* [103] adopts **FreMEn** [43] within an exploration scheme, where the planner predicts which areas are more likely to change at a certain time and generate the subsequent locations to explore. The experimental results showed that considering the environment dynamics increases the amount of information gathered compared to static models. Unlike **FreMEn** [43], Wang *et. al.* [44] models both aperiodic and periodic changes by an **ARMA** model. This model describes time series as stationary stochastic processes in terms of polynomials. While **FreMEn** [43] is able to update recursively its model online, **ARMA** [44] only is updated once a day based on past observations. However, the model [44] achieved a higher prediction accuracy than **FreMEn** [43] and a lower localization failure rate than both **FreMEn** [43] and Tipaldi *et. al.* [41].

Instead of modeling the dynamics in the map, Thomas *et. al.* [80] uses a KPConv network to predict online dynamic motion labels of points with single 3D laser scans as input. The method [80] is a self-supervised learning approach with two main modules: PointMap and PointRay. PointMap is an **ICP**-based **SLAM** algorithm to

provide a point cloud map for the annotation process. PointRay uses a similar approach to Pomerleau *et. al.* [92] to approximate ray tracing using spherical coordinates for obtaining the training annotation of dynamic labels: permanent (static points over all sessions), ground (to avoid ray tracing ground samples), and long-term (still objects in single sessions but relocated between sessions) and short-term (dynamic objects) movables, with the localization not considering the latter two. In the simulation experiments, PointMap with the proposed prediction module led to lower localization pose errors than the standard MCL algorithm.

2.4 Conclusions

The systematic review on long-term localization and mapping (available in Appendix A) discusses literature works on the topics of appearance invariance to changing conditions, dynamics modeling, map sparsification, and multi-session, among other computational issues, allowing the identification of current challenges in the literature and future directions for long-term SLAM. Given the focus of this research proposal on SLAM in dynamic environments, the current chapter focuses on literature works related to dynamics modeling found following the systematic methodology described in Appendix A and summarized in Section 2.1.

Next, the literature analysis of dynamics-related works presented in Section 2.3 discusses map representations proposed in the methodologies for modeling dynamics, dynamics identification and modeling using map matching, semantic detection of dynamic objects, and prediction of the environment dynamics. The environment representations proposed in the literature to handle moving elements in the scene are STM/LTM ([5, 30, 31]) memories and static/dynamic maps ([24, 34, 35]), and a multi-hypothesis framework ([78]). The first two techniques are very similar. Indeed, the LTM memory tries to filter dynamic data in the mapping process to only keep the most persistent and static observations in the map, similar to a static map. As for STM, this memory only keeps temporary and dynamic observations, equivalent to a dynamic representation of the environment. However, these map representations formulate assumptions on the environment dynamics, such as the number of consecutive observations to be considered a static observation ([30, 31]), certain semantic classes assumed to be dynamic observations ([34, 35]), or discrete timescales considered in the mapping process ([5]), or are susceptible to false positives and negatives ([24]), decreasing the accuracy of the environment representation. As for a multi-hypothesis formulation, while allowing recoverability from unmatched observations to the map, the limited computational resources on an AMR restrain the number of different hypotheses; plus, the proposed framework in [78] is very dependent on the uniqueness of the observations' signatures, which could not be valid for wide spaces and several rooms with similar appearance/structure. In terms of map matching, ray tracing-based techniques ([91–93]) exploit the free space information to identify dynamic changes between different observations. However, these techniques formulate a binary classification problem (static or dynamic) and are not suitable for low dynamics (also known as semi-static, i.e., changes such as structural or non-temporary ones), the latter due to assuming right way a change as a dynamic observation. Algorithms focused on the differences between the observations between the sensor data and the map ([25, 77, 81, 85–87]) are also focused only on the binary classification problem to reject dynamic changes for the localization process. Although the hybrid representation of NDT with occupancy grid maps ([94–96]) is required to model free space compared to the standard NDT formulation, the models used in those works favor newer observations, limiting the environment dynamics perceived by the algorithms.

Furthermore, the dynamic objects detection algorithms found in the literature focus on retrieving semantic data from the sensor observations, independently of being bounding box detection and classification ([35, 37,

101, 102]), or point-cloud ([97]) and image pixel-wise segmentation ([34, 38–40, 79, 84]). The main drawback of object detection-based evaluation of the environment dynamics is the prior assumption of certain classes being dynamic. This assumption may not stand valid in the long-term, *e.g.*, when an object presumably dynamic remains static (and could be used for localization temporarily) or, on the contrary, a presumably static element is moving in the scene. As for prediction of environment dynamics, those algorithms use mostly Markov-based models ([41, 42]), evaluate and predict the periodicity of changes ([43, 44, 103]), or use CNN-based prediction ([80]). Markov-based algorithms estimate the expected retention time for each cell of a 2D grid, inferring how often and for how long a dynamic observation is predicted to appear in the cells. However, this inference is based only on a model computed offline (based on previous measurements) due to the computational requirements of estimating the models, not integrating the most recent sensor observations. Although ARMA [44] improves over FreMEEn [43, 103], the former only updates offline its prediction model. Finally, the CNN-based approach [80] assumes a movable dynamic motion label for the points of single 3D LiDAR scans that could still be considered in the localization while in its temporary static state. Also, the learning process is only executed offline, even though uses ray-tracing for data labeling in a self-supervised way.

Overall, there are still open challenges in the literature on long-term SLAM in dynamic environments. Even the definition of dynamic itself, in terms of observations of the perceived environment by a mobile robot, is not clear in the literature. Indeed, what defines a dynamic, a semi-static, or a static observation? So, based on the literature discussion presented in this chapter and also on the systematic review available in Appendix A, the following items are considered to be open challenges for handling dynamic elements in the scene:

- **heterogeneous sensor configurations:** current literature on long-term SLAM focuses mostly on a single type of sensors and/or data;
- **multi-modal maps:** currently, only a single type of map (*e.g.*, grid, point cloud, feature) is considered in the mapping process when handling moving elements in the scene;
- **localization and mapping tightly-coupled with dynamics estimation:** instead of executing those processes independently, perform a joint estimation for the three processes;
- **online estimation of the observations dinamicity:** online determination of a dynamics value for the sensor observations;
- **human-machine interfaces:** interaction with users of AMRs, *e.g.*, for providing prior information provided by the user on the environment dynamics for localization and mapping tasks.

Lastly, the problem of lifelong localization and mapping in dynamic environments requires a SLAM framework. From the ones discussed in Section 2.2, GraphSLAM seems to obtain the best mapping accuracy, compared to scan matching or particle filter-based formulations [67], and to be the one most used recently, as noted in the data extraction process of the systematic review on long-term SLAM (see Appendix A). However, further discussion will be presented in the following chapters on how and which SLAM frameworks this research proposal will consider for its research plan. Also, all implementations mentioned in Section 2.2 (HectorSLAM [57], GMapping [66], and SLAM Toolbox [73]) are open-source and available as ROS [74, 75] packages (`hector_slam`¹, `gmapping`², and `slam_toolbox`³, respectively), facilitating possible testing and comparison during the execution of the research plan.

¹https://wiki.ros.org/hector_slam

²<https://wiki.ros.org/gmapping>

³https://wiki.ros.org/slam_toolbox

Chapter 3

Thesis Statement

3.1 Problem Statement

3.2 Research Questions and Hypotheses

3.3 Aim and Scope

3.4 Summary

Chapter 4

Preliminary Developments

4.1 Robot Calibration

4.2 Introduction to SLAM

4.3 Summary

Chapter 5

Research Plan

5.1 Tasks and Milestones

5.2 Timeline

5.3 Scientific Dissemination

5.4 Summary

References

- [1] Intel. *Autonomous Mobile Robot (AMR) overview: types and use cases*. Last accessed on Oct. 10, 2022. Sept. 2022. URL: <https://www.intel.com/content/www/us/en/robotics/autonomous-mobile-robots/overview.html>.
- [2] AgileIntel Research. *Global autonomous mobile robot market – trends, COVID-19 impact and growth forecasts to 2029*. Tech. rep. AgileIntel Research, May 2022. URL: <https://www.agileintelresearch.com/reportdetails/Global-Autonomous-Mobile-Robot-Market/43>.
- [3] Roland Siegwart, Illah R. Nourbakhsh, and Davide Scaramuzza. *Introduction to autonomous mobile robots*. Ed. by Ronald C. Arkin. Second. The MIT Press, 2004. ISBN: 978-0-262-01535-6.
- [4] Cesar Cadena, Luca Carlone, Henry Carrillo, Yasir Latif, Davide Scaramuzza, José Neira, Ian Reid, and John J. Leonard. “Past, present, and future of simultaneous localization and mapping: toward the robust-perception age”. In: *IEEE Transactions on Robotics* 32.6 (2016), pp. 1309–1332. DOI: [10.1109/TRO.2016.2624754](https://doi.org/10.1109/TRO.2016.2624754).
- [5] Peter Biber and Tom Duckett. “Experimental analysis of sample-based maps for long-term SLAM”. In: *International Journal of Robotics Research* 28.1 (2009), pp. 20–33. DOI: [10.1177/0278364908096286](https://doi.org/10.1177/0278364908096286).
- [6] Henrik Kretschmar and Cyrill Stachniss. “Information-theoretic compression of pose graphs for laser-based SLAM”. In: *International Journal of Robotics Research* 31.11 (2012), pp. 1219–1230. DOI: [10.1177/0278364912455072](https://doi.org/10.1177/0278364912455072).
- [7] Tsukamoto Taisho and Tanaka Kanji. “Mining DCNN landmarks for long-term visual SLAM”. In: 2016, pp. 570–576. DOI: [10.1109/ROBIO.2016.7866383](https://doi.org/10.1109/ROBIO.2016.7866383).
- [8] Zhe Xin, Xiaoguang Cui, Jixiang Zhang, Yiping Yang, and Yanqing Wang. “Visual place recognition with CNNs: from global to partial”. In: 2017, 6pp.–. DOI: [10.1109/IPTA.2017.8310121](https://doi.org/10.1109/IPTA.2017.8310121).
- [9] Zetao Chen, Lingqiao Liu, Inkyu Sa, Zongyuan Ge, and Margarita Chli. “Learning Context Flexible Attention Model for Long-Term Visual Place Recognition”. In: *IEEE Robotics and Automation Letters* 3.4 (2018), pp. 4015–4022. DOI: [10.1109/LRA.2018.2859916](https://doi.org/10.1109/LRA.2018.2859916).
- [10] Chao Yu, ZuXin Liu, Xin-Jun Liu, Fei Qiao, Yu Wang, Fugui Xie, Qi Wei, and Yi Yang. “A DenseNet feature-based loop closure method for visual SLAM system”. In: 2019, pp. 258–265. DOI: [10.1109/ROBIO49542.2019.8961714](https://doi.org/10.1109/ROBIO49542.2019.8961714).
- [11] Luis G. Camara, Carl Gäbert, and Libor Pěručil. “Highly Robust Visual Place Recognition Through Spatial Matching of CNN Features”. In: 2020, pp. 3748–3755. DOI: [10.1109/ICRA40945.2020.9196967](https://doi.org/10.1109/ICRA40945.2020.9196967).

- [12] Bingxi Liu, Fulin Tang, Yujie Fu, Yanqun Yang, and Yihong Wu. “A Flexible and Efficient Loop Closure Detection Based on Motion Knowledge”. In: 2021, pp. 11241–7. DOI: [10.1109/ICRA48506.2021.9561126](https://doi.org/10.1109/ICRA48506.2021.9561126).
- [13] Peng Yin, Lingyun Xu, Zhe Liu, Lu Li, Hadi Salman, Yuqing He, Weiliang Xu, Hesheng Wang, and Howie Choset. “Stabilize an Unsupervised Feature Learning for LiDAR-based Place Recognition”. In: 2018, pp. 1162–1167. DOI: [10.1109/IROS.2018.8593562](https://doi.org/10.1109/IROS.2018.8593562).
- [14] Giseop Kim, Byungjae Park, and Ayoung Kim. “1-Day Learning, 1-Year Localization: Long-Term LiDAR Localization Using Scan Context Image”. In: *IEEE Robotics and Automation Letters* 4.2 (2019), pp. 1948–1955. DOI: [10.1109/LRA.2019.2897340](https://doi.org/10.1109/LRA.2019.2897340).
- [15] Xuecheng Xu, Huan Yin, Zexi Chen, Yuehua Li, Yue Wang, and Rong Xiong. “DiSCO: Differentiable Scan Context with Orientation”. In: *IEEE Robotics and Automation Letters* 6.2 (2021), pp. 2791–2798. DOI: [10.1109/LRA.2021.3060741](https://doi.org/10.1109/LRA.2021.3060741).
- [16] Huan Yin, Xuecheng Xu, Yue Wang, and Rong Xiong. “Radar-to-Lidar: Heterogeneous Place Recognition via Joint Learning”. In: *Frontiers in Robotics and AI* 8 (2021), p. 661199. DOI: [10.3389/frobt.2021.661199](https://doi.org/10.3389/frobt.2021.661199).
- [17] Daniele De Martini, Matthew Gadd, and Paul Newman. “kRadar++: Coarse-to-fine FMCW scanning radar localisation”. In: *Sensors* 20.21 (2020), pp. 1–23. DOI: [10.3390/s20216002](https://doi.org/10.3390/s20216002).
- [18] Julie Stephany Berrio, James Ward, Stewart Worrall, and Eduardo Nebot. “Identifying robust landmarks in feature-based maps”. In: vol. 2019-June. 2019, pp. 1166–1172. DOI: [10.1109/IVS.2019.8814289](https://doi.org/10.1109/IVS.2019.8814289).
- [19] Julie Stephany Berrio, Stewart Worrall, M. Shan, and Eduardo Nebot. “Long-Term Map Maintenance Pipeline for Autonomous Vehicles”. In: *IEEE Transactions on Intelligent Transportation Systems* (2021). DOI: [10.1109/TITS.2021.3094485](https://doi.org/10.1109/TITS.2021.3094485).
- [20] Henrik Kretschmar, Giorgio Grisetti, and Cyrill Stachniss. “Lifelong Map Learning for Graph-based SLAM in Static Environments”. In: *KI - Kunstliche Intelligenz* 24.3 (2010), pp. 199–206. DOI: [10.1007/s13218-010-0034-2](https://doi.org/10.1007/s13218-010-0034-2).
- [21] Guoquan Huang, Michael Kaess, and John J. Leonard. “Consistent sparsification for graph optimization”. In: 2013, pp. 150–157. DOI: [10.1109/ECMR.2013.6698835](https://doi.org/10.1109/ECMR.2013.6698835).
- [22] Nicholas Carlevaris-Bianco, Michael Kaess, and Ryan M. Eustice. “Generic node removal for factor-graph SLAM”. In: *IEEE Transactions on Robotics* 30.6 (2014), pp. 1371–1385. DOI: [10.1109/TRO.2014.2347571](https://doi.org/10.1109/TRO.2014.2347571).
- [23] Mladen Mazuran, Wolfram Burgard, and Gian Diego Tipaldi. “Nonlinear factor recovery for long-term SLAM”. In: *International Journal of Robotics Research* 35.1-3 (2016), pp. 50–72. DOI: [10.1177/0278364915581629](https://doi.org/10.1177/0278364915581629).
- [24] Aisha Walcott-Bryant, Michael Kaess, Hordur Johannsson, and John J. Leonard. “Dynamic pose graph SLAM: Long-term mapping in low dynamic environments”. In: 2012, pp. 1871–1878. DOI: [10.1109/IROS.2012.6385561](https://doi.org/10.1109/IROS.2012.6385561).
- [25] Federico Boniardi, Tim Caselitz, Rainer Kümmerle, and Wolfram Burgard. “A pose graph-based localization system for long-term navigation in CAD floor plans”. In: *Robotics and Autonomous Systems* 112 (2019), pp. 84–97. DOI: [10.1016/j.robot.2018.11.003](https://doi.org/10.1016/j.robot.2018.11.003).

- [26] Hordur Johannsson, Michael Kaess, Maurice Fallon, and John J. Leonard. “Temporally scalable visual SLAM using a reduced pose graph”. In: 2013, pp. 54–61. DOI: [10.1109/ICRA.2013.6630556](https://doi.org/10.1109/ICRA.2013.6630556).
- [27] Taiping Zeng and Bailu Si. “A brain-inspired compact cognitive mapping system”. In: *Cognitive Neurodynamics* 15.1 (2021), pp. 91–101. DOI: [10.1007/s11571-020-09621-6](https://doi.org/10.1007/s11571-020-09621-6).
- [28] Gerhard Kurz, Matthias Holoch, and Peter Biber. “Geometry-based Graph Pruning for Lifelong SLAM”. In: 2021, pp. 3313–3320. DOI: [10.1109/IROS51168.2021.9636530](https://doi.org/10.1109/IROS51168.2021.9636530).
- [29] Waqas Ali, Peilin Liu, Rendong Ying, and Zheng Gong. “A Life-Long SLAM Approach Using Adaptable Local Maps Based on Rasterized LIDAR Images”. In: *IEEE Sensors Journal* 21.19 (2021), pp. 21740–21749. DOI: [10.1109/JSEN.2021.3100882](https://doi.org/10.1109/JSEN.2021.3100882).
- [30] Feras Dayoub, Grzegorz Cielniak, and Tom Duckett. “Long-term experiments with an adaptive spherical view representation for navigation in changing environments”. In: *Robotics and Autonomous Systems* 59.5 (2011), pp. 285–295. DOI: [10.1016/j.robot.2011.02.013](https://doi.org/10.1016/j.robot.2011.02.013).
- [31] Bladimir Bacca, Joaquín Salví, and Xavier Cufí. “Long-term mapping and localization using feature stability histograms”. In: *Robotics and Autonomous Systems* 61.12 (2013), pp. 1539–58. DOI: [10.1016/j.robot.2013.07.003](https://doi.org/10.1016/j.robot.2013.07.003).
- [32] Marcin Dymczyk, Thomas Schneider, Igor Gilitschenski, Roland Siegwart, and Elena Stumm. “Erasing bad memories: Agent-side summarization for long-term mapping”. In: vol. 2016-November. 2016, pp. 4572–4579. DOI: [10.1109/IROS.2016.7759673](https://doi.org/10.1109/IROS.2016.7759673).
- [33] Peter Mühlfellner, Mathias Bürki, Michael Bosse, Wojciech Derendarz, Roland Philippsen, and Paul Furgale. “Summary Maps for Lifelong Visual Localization”. In: *Journal of Field Robotics* 33.5 (2016), pp. 561–590. DOI: [10.1002/rob.21595](https://doi.org/10.1002/rob.21595).
- [34] Kai Wang, Yimin Lin, Luowei Wang, Liming Han, Minjie Hua, Xiang Wang, Shiguo Lian, and Bill Huang. “A unified framework for mutual improvement of SLAM and semantic segmentation”. In: vol. 2019-May. 2019, pp. 5224–5230. DOI: [10.1109/ICRA.2019.8793499](https://doi.org/10.1109/ICRA.2019.8793499).
- [35] Shifan Zhu, Xinyu Zhang, Shichun Guo, Jun Li, and Huaping Liu. “Lifelong Localization in Semi-Dynamic Environment”. In: vol. 2021-May. 2021, pp. 14389–14395. DOI: [10.1109/ICRA48506.2021.9561584](https://doi.org/10.1109/ICRA48506.2021.9561584).
- [36] R.C. Atkinson and R.M. Shiffrin. “Human Memory: A Proposed System and its Control Processes”. In: ed. by Kenneth W. Spence and Janet Taylor Spence. Vol. 2. Psychology of Learning and Motivation. Academic Press, 1968, pp. 89–195. DOI: [https://doi.org/10.1016/S0079-7421\(08\)60422-3](https://doi.org/10.1016/S0079-7421(08)60422-3).
- [37] Yufeng Yue, Chule Yang, Jun Zhang, Mingxing Wen, Zhenyu Wu, Haoyuan Zhang, and Danwei Wang. “Day and Night Collaborative Dynamic Mapping in Unstructured Environment Based on Multimodal Sensors”. In: 2020, pp. 2981–7. DOI: [10.1109/ICRA40945.2020.9197072](https://doi.org/10.1109/ICRA40945.2020.9197072).
- [38] Gaurav Singh, Meiqing Wu, Siew-Kei Lam, and Do Van Minh. “Hierarchical Loop Closure Detection for Long-term Visual SLAM with Semantic-Geometric Descriptors”. In: vol. 2021-September. 2021, pp. 2909–2916. DOI: [10.1109/ITSC48978.2021.9564866](https://doi.org/10.1109/ITSC48978.2021.9564866).
- [39] Yafei Song, Di Zhu, Jia Li, Yonghong Tian, and Mingyang Li. “Learning Local Feature Descriptor with Motion Attribute For Vision-based Localization”. In: 2019, pp. 3794–801. DOI: [10.1109/IROS40897.2019.8967749](https://doi.org/10.1109/IROS40897.2019.8967749).

- [40] Pranav Ganti and Steven L. Waslander. “Network uncertainty informed semantic feature selection for visual SLAM”. In: 2019, pp. 121–128. DOI: [10.1109/CRV.2019.00024](https://doi.org/10.1109/CRV.2019.00024).
- [41] Gian Diego Tipaldi, Daniel Meyer-Delius, and Wolfram Burgard. “Lifelong localization in changing environments”. In: *International Journal of Robotics Research* 32.14 (2013), pp. 1662–1678. DOI: [10.1177/0278364913502830](https://doi.org/10.1177/0278364913502830).
- [42] Matthias Rapp, Markus Hahn, Markus Thom, Jürgen Dickmann, and Klaus Dietmayer. “Semi-Markov Process Based Localization using radar in Dynamic Environments”. In: *2015 IEEE International Intelligent Transportation Systems Conference (ITSC)*. 2015, pp. 423–429. DOI: [10.1109/ITSC.2015.77](https://doi.org/10.1109/ITSC.2015.77).
- [43] Tomáš Krajník, Jaime Pulido Fentanes, João Machado Santos, and Tom Duckett. “FreMEn: Frequency map enhancement for long-term mobile robot autonomy in changing environments”. In: *IEEE Transactions on Robotics* 33.4 (2017), pp. 964–977. DOI: [10.1109/TRO.2017.2665664](https://doi.org/10.1109/TRO.2017.2665664).
- [44] Lisai Wang, Weidong Chen, and Jingchuan Wang. “Long-term localization with time series map prediction for mobile robots in dynamic environments”. In: vol. 2020-January. 2020, pp. 8587–8593. DOI: [10.1109/IROS45743.2020.9468884](https://doi.org/10.1109/IROS45743.2020.9468884).
- [45] Thomas Hahn, Morten Irgens, Bernd Liepert, Barry O’Sullivan, Arnold WM Smeulders, David Bisset, Edward Curry, Michela Milano, and Sonja Zillner. *Strategic Research, Innovation and Deployment Agenda (SRIDA)*. Tech. rep. Big Data Value Association (BDVA), Confederation of Laboratories for Artificial Intelligence Research in Europe (CLAIRE), European Lab for Learning & Intelligent Systems (ELLIS), European Association for Artificial Intelligence (EurAI), and euRobotics, Sept. 2020. URL: <https://adr-association.eu/wp-content/uploads/2020/09/AI-Data-Robotics-Partnership-SRIDA-V3.0-1.pdf>.
- [46] United Nations. *Transforming our world: the 2030 agenda for sustainable development*. Last accessed on Oct. 17, 2022. Sept. 2015. URL: <https://sdgs.un.org/publications/transforming-our-world-2030-agenda-sustainable-development-17981>.
- [47] Ken Peffers, Tuure Tuunanen, Marcus A. Rothenberger, and Samir Chatterjee. “A design science research methodology for information systems research”. In: *Journal of Management Information Systems* 24.3 (2007), pp. 45–77. DOI: [10.2753/MIS0742-1222240302](https://doi.org/10.2753/MIS0742-1222240302).
- [48] Matthew J Page, David Moher, Patrick M Bossuyt, Isabelle Boutron, Tammy C Hoffmann, Cynthia D Mulrow, Larissa Shamseer, Jennifer M Tetzlaff, Elie A Akl, Sue E Brennan, Roger Chou, Julie Glanville, Jeremy M Grimshaw, Asbjørn Hróbjartsson, Manoj M Lalu, Tianjing Li, Elizabeth W Loder, Evan Mayo-Wilson, Steve McDonald, Luke A McGuinness, Lesley A Stewart, James Thomas, Andrea C Tricco, Vivian A Welch, Penny Whiting, and Joanne E McKenzie. “PRISMA 2020 explanation and elaboration: updated guidance and exemplars for reporting systematic reviews”. In: *BMJ* 372.n160 (Mar. 2021). DOI: [10.1136/bmj.n160](https://doi.org/10.1136/bmj.n160).
- [49] Association for Computing Machinery (ACM). *ACM Digital Library*. Last accessed on Oct. 25, 2022. Jan. 2020. URL: <https://dl.acm.org/>.
- [50] Digital Science. *Dimensions*. Last accessed on Oct. 25, 2022. 2018. URL: <https://www.dimensions.ai/>.
- [51] IEEE. *IEEE Xplore*. Last accessed on Oct. 25, 2022. URL: <https://ieeexplore.ieee.org/>.

- [52] Engineering Village. *INSPEC*. Last accessed on Oct. 25, 2022. URL: <https://www.engineeringvillage.com/search/quick.url?database=2>.
- [53] Elsevier. *Scopus*. Last accessed on Oct. 25, 2022. URL: <https://www.scopus.com/>.
- [54] Clarivate. *Web of Science*. Last accessed on Oct. 25, 2022. URL: <https://www.webofscience.com/>.
- [55] Vitor Freitas. *Parsifal*. Last accessed on May 12, 2022. 2014. URL: <https://parsif.al/>.
- [56] S. Thrun, W. Burgard, and D. Fox. *Probabilistic robotics*. The MIT Press, 2005. ISBN: 978-0-262-20162-9.
- [57] Stefan Kohlbrecher, Oskar von Stryk, Johannes Meyer, and Uwe Klingauf. “A flexible and scalable SLAM system with full 3D motion estimation”. In: *2011 IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR)*. 2011, pp. 155–160. DOI: [10.1109/SSRR.2011.6106777](https://doi.org/10.1109/SSRR.2011.6106777).
- [58] Martin Lauer, Sascha Lange, and Martin Riedmiller. “Calculating the Perfect Match: An Efficient and Accurate Approach for Robot Self-localization”. In: *RoboCup 2005: Robot Soccer World Cup IX*. 2006, pp. 142–153. DOI: [10.1007/11780519_13](https://doi.org/10.1007/11780519_13).
- [59] P.J. Besl and Neil D. McKay. “A method for registration of 3-D shapes”. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 14.2 (1992), pp. 239–256. DOI: [10.1109/34.121791](https://doi.org/10.1109/34.121791).
- [60] Jacopo Serafin and Giorgio Grisetti. “NICP: Dense normal based point cloud registration”. In: *2015 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2015, pp. 742–749. DOI: [10.1109/IROS.2015.7353455](https://doi.org/10.1109/IROS.2015.7353455).
- [61] P. Biber and W. Strasser. “The normal distributions transform: a new approach to laser scan matching”. In: *2003 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2003, 10.1109/IROS.2003.1249285. DOI: [2743--2748](https://doi.org/10.1109/IROS.2003.1249285).
- [62] Arnaud Doucet, Nando de Freitas, Kevin Murphy, and Stuart Russell. “Rao-Blackwellised Filtering for Dynamic Bayesian Networks”. In: *Proceedings of the 16th Conference on Uncertainty in Artificial Intelligence (UAI-00)*. 2000, p. 2000. URL: https://www.cs.ubc.ca/~murphyk/Papers/rbpf_uai00.pdf.
- [63] Giorgio Grisetti, Rainer Kümmerle, Cyrill Stachniss, and Wolfram Burgard. “A tutorial on graph-based SLAM”. In: *IEEE Intelligent Transportation Systems Magazine* 2.4 (2010), pp. 31–43. DOI: [10.1109/MITS.2010.939925](https://doi.org/10.1109/MITS.2010.939925).
- [64] Michael Montemerlo, Sebastian Thrun, Daphne Koller, and Ben Wegbreit. “FastSLAM: A Factored Solution to the Simultaneous Localization and Mapping Problem”. In: *Proceedings of the Eighteenth National Conference on Artificial Intelligence (AAAI-02)*. 2002, pp. 593–598. URL: <https://aaai.org/Papers/AAAI/2002/AAAI02-089.pdf>.
- [65] Mike Montemerlo, Sebastian Thrun, Daphne Koller, and Ben Wegbreit. “FastSLAM 2.0: An Improved Particle Filtering Algorithm for Simultaneous Localization and Mapping that Provably Converges”. In: *Proceedings of the Eighteenth International Joint Conference on Artificial Intelligence (IJCAI-03)*. 2003, pp. 1151–1156. URL: <https://www.ijcai.org/Proceedings/03/Papers/165.pdf>.
- [66] Giorgio Grisetti, Cyrill Stachniss, and Wolfram Burgard. “Improved Techniques for Grid Mapping With Rao-Blackwellized Particle Filters”. In: *IEEE Transactions on Robotics* 23.1 (2007), pp. 34–46. DOI: [10.1109/TRO.2006.889486](https://doi.org/10.1109/TRO.2006.889486).

- [67] Guillaume Bresson, Zayed Alsayed, Li Yu, and Sébastien Glaser. “Simultaneous Localization and Mapping: A Survey of Current Trends in Autonomous Driving”. In: *IEEE Transactions on Intelligent Vehicles* 2.3 (2017), pp. 194–220. DOI: [10.1109/TIV.2017.2749181](https://doi.org/10.1109/TIV.2017.2749181).
- [68] Giorgio Grisetti, Cyrill Stachniss, and Wolfram Burgard. “Nonlinear Constraint Network Optimization for Efficient Map Learning”. In: *IEEE Transactions on Intelligent Transportation Systems* 10.3 (2009), pp. 428–439. DOI: [10.1109/TITS.2009.2026444](https://doi.org/10.1109/TITS.2009.2026444).
- [69] Michael Kaess, Ananth Ranganathan, and Frank Dellaert. “iSAM: Incremental Smoothing and Mapping”. In: *IEEE Transactions on Robotics* 24.6 (2008), pp. 1365–1378. DOI: [10.1109/TRO.2008.2006706](https://doi.org/10.1109/TRO.2008.2006706).
- [70] Michael Kaess, Hordur Johannsson, Richard Roberts, Viorela Ila, John Leonard, and Frank Dellaert. “iSAM2: Incremental smoothing and mapping with fluid relinearization and incremental variable re-ordering”. In: 2011, pp. 3281–3288. DOI: [10.1109/ICRA.2011.5979641](https://doi.org/10.1109/ICRA.2011.5979641).
- [71] Rainer Kümmerle, Giorgio Grisetti, Hauke Strasdat, Kurt Konolige, and Wolfram Burgard. “G2o: A general framework for graph optimization”. In: 2011, pp. 3607–3613. DOI: [10.1109/ICRA.2011.5979949](https://doi.org/10.1109/ICRA.2011.5979949).
- [72] Richard J. Rossi. *Mathematical Statistics: An Introduction to Likelihood Based Inference*. First. John Wiley & Sons, 2018. ISBN: 978-1-118-77104-4.
- [73] Steve Macenski and Ivona Jambrecic. “SLAM Toolbox: SLAM for the dynamic world”. In: *Journal of Open Source Software* 6.61 (2021), p. 2783. DOI: [10.21105/joss.02783](https://doi.org/10.21105/joss.02783).
- [74] Open Robotics. *ROS: Robot Operating System*. Last accessed on Nov. 22, 2022. 2021. URL: <https://www.ros.org/>.
- [75] Open Robotics. *ROS 2 Documentation: Foxy*. Last accessed on Nov. 22, 2022. 2022. URL: <https://docs.ros.org/en/foxy/index.html>.
- [76] Sameer Agarwal, Keir Mierle, and The Ceres Solver Team. *Ceres Solver*. Last accessed on Nov. 22, 2022. Mar. 2010. URL: <https://github.com/ceres-solver/ceres-solver>.
- [77] Chenxin Liu, Jiahui Qin, Shuai Wang, Lei Yu, and Yaonan Wang. “Accurate RGB-D SLAM in dynamic environments based on dynamic visual feature removal”. In: *Science China Information Sciences* 65.10 (2022), p. 202206. DOI: [10.1007/s11432-021-3425-8](https://doi.org/10.1007/s11432-021-3425-8).
- [78] Kristin Nielsen and Gustaf Hendeby. “Feature Based Multi-Hypothesis Map Representation for Localization in Non-Static Environments”. In: 2022, pp. 1–8. DOI: [10.23919/FUSION49751.2022.9841255](https://doi.org/10.23919/FUSION49751.2022.9841255).
- [79] Ruibin Guo and Xinghua Liu. “Ground Enhanced RGB-D SLAM for Dynamic Environments”. In: 2021, pp. 1171–1177. DOI: [10.1109/ROBIO54168.2021.9739362](https://doi.org/10.1109/ROBIO54168.2021.9739362).
- [80] Hugues Thomas, Matthieu Gallet de Saint Aurin, Jian Zhang, and Timothy D. Barfoot. “Learning Spatiotemporal Occupancy Grid Maps for Lifelong Navigation in Dynamic Scenes”. In: 2022, pp. 484–490. DOI: [10.1109/ICRA46639.2022.9812297](https://doi.org/10.1109/ICRA46639.2022.9812297).
- [81] Jiahui Fu, Chengyuan Lin, Yuichi Taguchi, Andrea Cohen, Yifu Zhang, Stephen Mylabathula, and John J. Leonard. “PlaneSDF-Based Change Detection for Long-Term Dense Mapping”. In: *IEEE Robotics and Automation Letters* 7.4 (2022), pp. 9667–9674. DOI: [10.1109/LRA.2022.3191794](https://doi.org/10.1109/LRA.2022.3191794).

- [82] Hugues Thomas, Ben Agro, Mona Gridseth, Jian Zhang, and Timothy D. Barfoot. “Self-Supervised Learning of Lidar Segmentation for Autonomous Indoor Navigation”. In: vol. 2021-May. 2021, pp. 14047–14053. DOI: [10.1109/ICRA48506.2021.9561701](https://doi.org/10.1109/ICRA48506.2021.9561701).
- [83] Raúl Mur-Artal and Juan D. Tardós. “ORB-SLAM2: An Open-Source SLAM System for Monocular, Stereo, and RGB-D Cameras”. In: *IEEE Transactions on Robotics* 33.5 (2017), pp. 1255–1262. DOI: [10.1109/TRO.2017.2705103](https://doi.org/10.1109/TRO.2017.2705103).
- [84] Berta Bescos, José M. Fácil, Javier Civera, and José Neira. “DynaSLAM: Tracking, Mapping, and Inpainting in Dynamic Scenes”. In: *IEEE Robotics and Automation Letters* 3.4 (2018), pp. 4076–4083. DOI: [10.1109/LRA.2018.2860039](https://doi.org/10.1109/LRA.2018.2860039).
- [85] Joydeep Biswas and Manuela M. Veloso. “Episodic non-Markov localization”. In: *Robotics and Autonomous Systems* 87 (2017), pp. 162–176. DOI: [10.1016/j.robot.2016.09.005](https://doi.org/10.1016/j.robot.2016.09.005).
- [86] Mingming Zhang, Yiming Chen, and Mingyang Li. “SDF-Loc: Signed distance Field based 2D relocalization and map update in dynamic environments”. In: vol. 2019-July. 2019, pp. 1997–2004. DOI: [10.23919/acc.2019.8814347](https://doi.org/10.23919/acc.2019.8814347).
- [87] Zheng-Jun Du, Shi-Sheng Huang, Tai-Jiang Mu, Qunhe Zhao, Ralph R. Martin, and Kun Xu. “Accurate Dynamic SLAM Using CRF-Based Long-Term Consistency”. In: *IEEE Transactions on Visualization and Computer Graphics* 28.4 (2022), pp. 1745–57. DOI: [10.1109/TVCG.2020.3028218](https://doi.org/10.1109/TVCG.2020.3028218).
- [88] Raúl Mur-Artal, J. M. M. Montiel, and Juan D. Tardós. “ORB-SLAM: A Versatile and Accurate Monocular SLAM System”. In: *IEEE Transactions on Robotics* 31.5 (2015), pp. 1147–1163. DOI: [10.1109/TRO.2015.2463671](https://doi.org/10.1109/TRO.2015.2463671).
- [89] Zhichen Pan, Haoyao Chen, Silin Li, and Yunhui Liu. “Clustermapping building and relocalization in urban environments for unmanned vehicles”. In: *Sensors* 19.19 (2019), p. 4252. DOI: [10.3390/s19194252](https://doi.org/10.3390/s19194252).
- [90] Xiaqing Ding, Yue Wang, Rong Xiong, Dongxuan Li, Li Tang, Huan Yin, and Liang Zhao. “Persistent Stereo Visual Localization on Cross-Modal Invariant Map”. In: *IEEE Transactions on Intelligent Transportation Systems* 21.11 (2020), pp. 4646–4658. DOI: [10.1109/TITS.2019.2942760](https://doi.org/10.1109/TITS.2019.2942760).
- [91] María T. Lázaro, Roberto Capobianco, and Giorgio Grisetti. “Efficient Long-term Mapping in Dynamic Environments”. In: 2018, pp. 153–160. DOI: [10.1109/IROS.2018.8594310](https://doi.org/10.1109/IROS.2018.8594310).
- [92] Francois Pomerleau, Philipp Krüsi, Francis Colas, Paul Furgale, and Roland Siegwart. “Long-term 3D map maintenance in dynamic environments”. In: 2014, pp. 3712–3719. DOI: [10.1109/ICRA.2014.6907397](https://doi.org/10.1109/ICRA.2014.6907397).
- [93] Su-Yong An, Lae-Kyoung Lee, and Se-Young Oh. “Ceiling vision-based active SLAM framework for dynamic and wide-open environments”. In: *Autonomous Robots* 40.2 (2016), pp. 291–324. DOI: [10.1007/s10514-015-9453-0](https://doi.org/10.1007/s10514-015-9453-0).
- [94] Erik Einhorn and Horst-Michael Gross. “Generic 2D/3D SLAM with NDT maps for lifelong application”. In: 2013, pp. 240–247. DOI: [10.1109/ECMR.2013.6698849](https://doi.org/10.1109/ECMR.2013.6698849).
- [95] Erik Einhorn and Horst-Michael Gross. “Generic NDT mapping in dynamic environments and its application for lifelong SLAM”. In: *Robotics and Autonomous Systems* 69.1 (2015), pp. 28–39. DOI: [10.1016/j.robot.2014.08.008](https://doi.org/10.1016/j.robot.2014.08.008).

- [96] Jari P. Saarinen, Henrik Andreasson, Todor Stoyanov, and Achim J. Lilienthal. “3D normal distributions transform occupancy maps: An efficient representation for mapping in dynamic environments”. In: *International Journal of Robotics Research* 32.14 (2013), pp. 1627–44. DOI: [10.1177/0278364913499415](https://doi.org/10.1177/0278364913499415).
- [97] L. Sun, Zhi Yan, Anestis Zaganidis, Cheng Zhao, and Tom Duckett. “Recurrent-OctoMap: Learning State-Based Map Refinement for Long-Term Semantic Mapping with 3-D-Lidar Data”. In: *IEEE Robotics and Automation Letters* 3.4 (2018), pp. 3749–3756. DOI: [10.1109/LRA.2018.2856268](https://doi.org/10.1109/LRA.2018.2856268).
- [98] Michael J. Milford and Gordon. F. Wyeth. “SeqSLAM: Visual route-based navigation for sunny summer days and stormy winter nights”. In: 2012, pp. 1643–1649. DOI: [10.1109/ICRA.2012.6224623](https://doi.org/10.1109/ICRA.2012.6224623).
- [99] Mark Cummins and Paul Newman. “FAB-MAP: Probabilistic Localization and Mapping in the Space of Appearance”. In: *International Journal of Robotics Research* 27.6 (2008), pp. 647–665. DOI: [10.1177/0278364908090961](https://doi.org/10.1177/0278364908090961).
- [100] Josef Sivic and Andrew Zisserman. “Video Google: a text retrieval approach to object matching in videos”. In: 2003, pp. 1470–1477. DOI: [10.1109/ICCV.2003.1238663](https://doi.org/10.1109/ICCV.2003.1238663).
- [101] Shiqiang Yang, Guohao Fan, Lele Bai, Cheng Zhao, and Dexin Li. “SGC-VSLAM: A semantic and geometric constraints VSLAM for dynamic indoor environments”. In: *Sensors* 20.8 (2020), p. 2432. DOI: [10.3390/s20082432](https://doi.org/10.3390/s20082432).
- [102] Zhiwei Xing, Xiaorui Zhu, and Dingcheng Dong. “DE-SLAM: SLAM for highly dynamic environment”. In: *Journal of Field Robotics* (2022). DOI: [10.1002/rob.22062](https://doi.org/10.1002/rob.22062).
- [103] João Machado Santos, Tomáš Krajník, Jaime Pulido Fentanes, and Tom Duckett. “Lifelong Information-Driven Exploration to Complete and Refine 4-D Spatio-Temporal Maps”. In: *IEEE Robotics and Automation Letters* 1.2 (2016), pp. 684–691. DOI: [10.1109/LRA.2016.2516594](https://doi.org/10.1109/LRA.2016.2516594).

Appendix A

A Systematic Literature Review on Long-Term Localization and Mapping for Mobile Robots (preprint)

Ricardo B. Sousa, Héber M. Sobreira, and António Paulo Moreira. “A Systematic Literature Review on Long-Term Localization and Mapping for Mobile Robots” [Manuscript submitted for publication]. 2022. Department of Electrical and Computer Engineering, Faculty of Engineering of the University of Porto (FEUP). Centre for Robotics in Industry and Intelligent Systems (CRIIS), INESC TEC – Institute for Systems and Computer Engineering, Technology and Science. URL: <https://github.com/sousarbarb/slr-ltln-mr/blob/main/preprint/pdf.pdf>